

Chapter 11.A

Molecular Shape and Bonding Theories

Chemical Principles I

CHEM 1211K

1. Use Lewis structures to predict molecular geometry using VSEPR.
2. Describe and apply VSEPR geometries and deviations from them.
3. Identify polar and nonpolar molecules using electronegativity values and molecular shape.
4. Explain valence bond theory.
5. Describe and recognize sp , sp^2 , and sp^3 hybridization based on the Lewis structure of a molecule.
6. Define, describe, and identify sigma and pi bonds in molecules.
7. Describe the limitations of valence bond theory.
8. Describe and apply the tenets of molecular orbital theory.
9. Determine bond order and magnetic properties using molecular orbital diagrams.

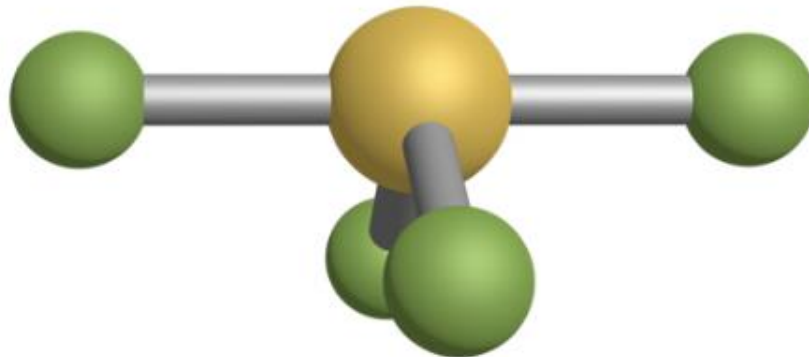
11.1 Valence Shell Electron Pair Repulsion (VSEPR) Theory



- How can we predict the three-dimensional geometry of a molecule from its Lewis structure?
- According to **valence shell electron pair repulsion (VSEPR) theory**, regions of electron density (**electron domains**) try to position themselves as far apart from one another as possible to minimize repulsion. Molecular shape follows from this concept.
- Magnitude of repulsion for bonding and nonbonding (lone pair) domains...

Determine the total number of electron domains, the number of bonding groups, and the number of nonbonding lone pairs for the molecule below. The yellow atom is sulfur and the green atoms are chlorines.

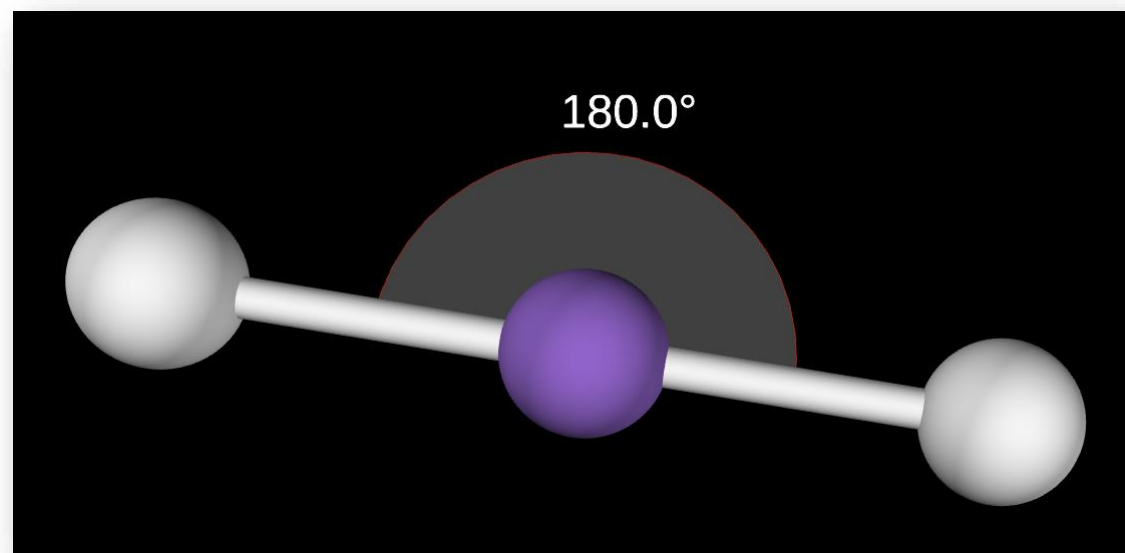
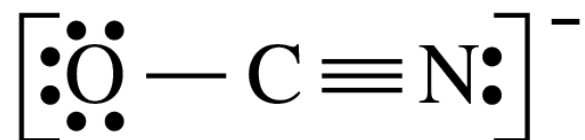
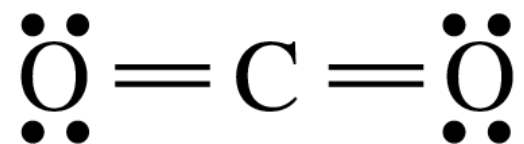
To answer this question, enter the number of electron domains at the central atom.



- **Molecular geometry** is the arrangement of *atoms* around the central atom (equivalently, bonding domains only).

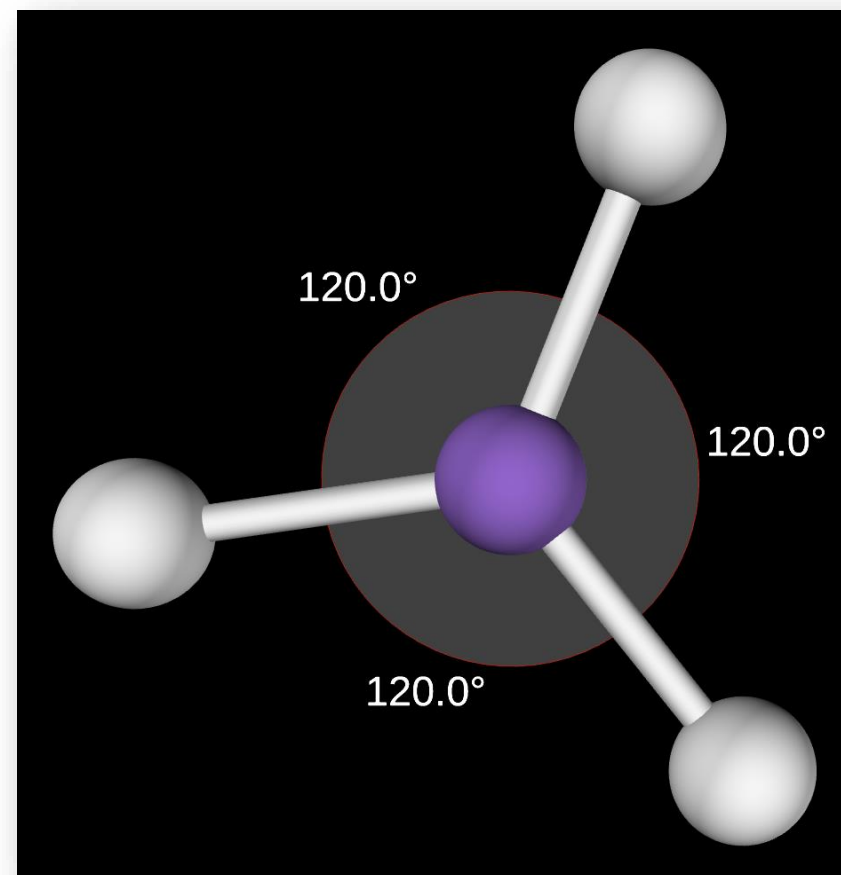
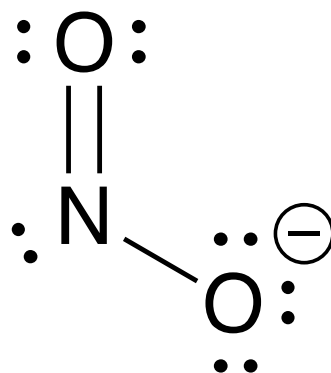
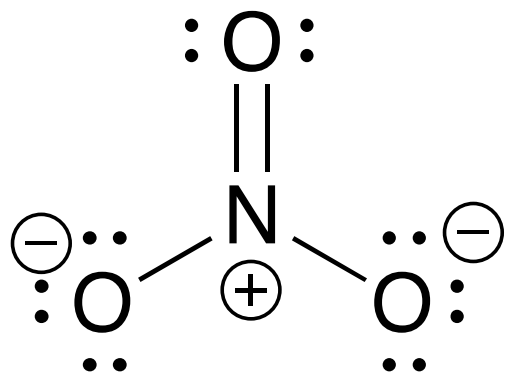
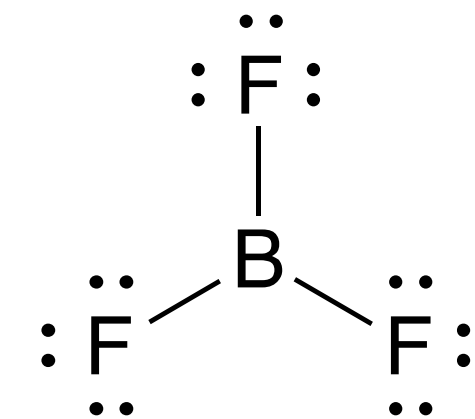
11.1 Linear Geometry

- Electron geometry follows from the **steric number** at the central atom.
https://phet.colorado.edu/sims/html/molecule-shapes/latest/molecule-shapes_all.html
- Linear** electron geometry (steric number 2)



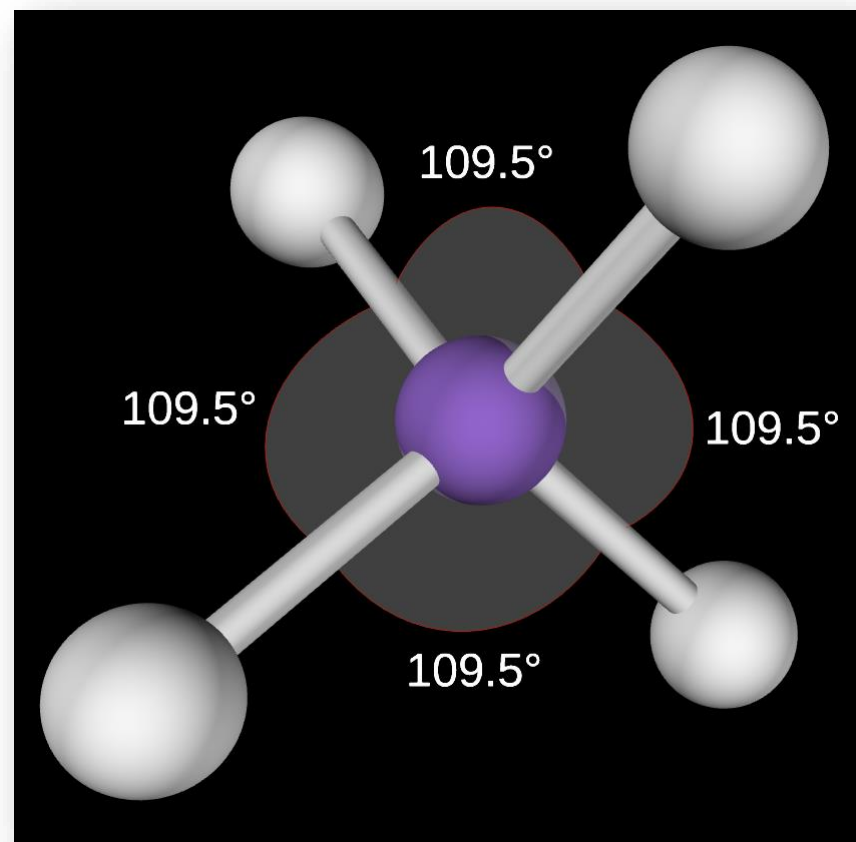
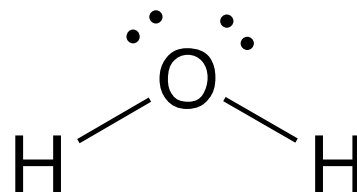
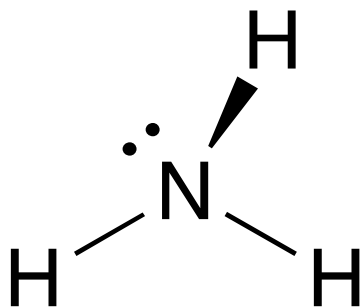
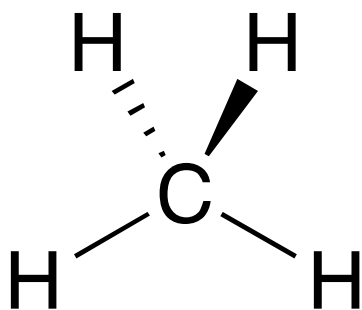
11.1 Trigonal Planar Geometry

- Electron geometry follows from the **steric number** at the central atom.
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- Trigonal planar** electron geometry (steric number 3)



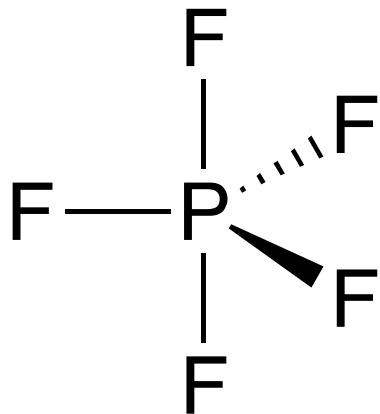
11.1 Tetrahedral Geometry

- Electron geometry follows from the **steric number** at the central atom.
https://phet.colorado.edu/sims/html/molecule-shapes/latest/molecule-shapes_all.html
- **Tetrahedral** electron geometry (steric number 4)

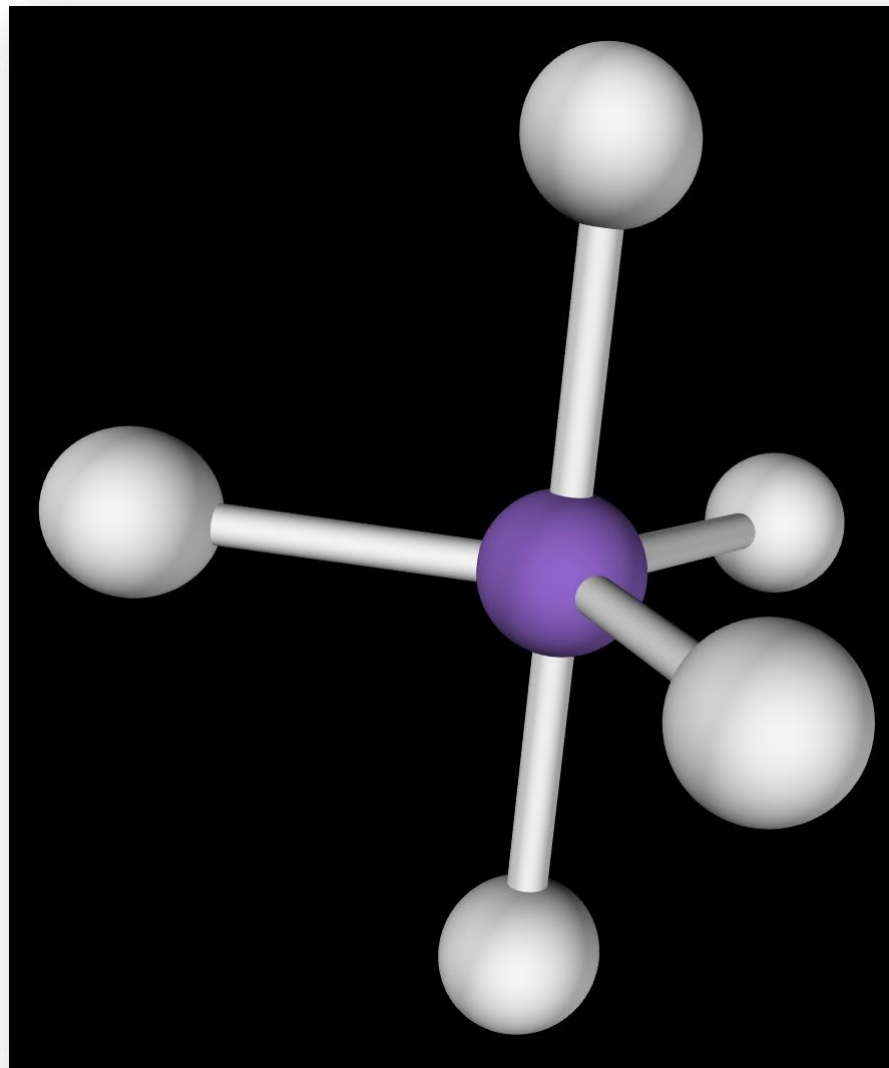


11.1 Trigonal Bipyramidal Geometry

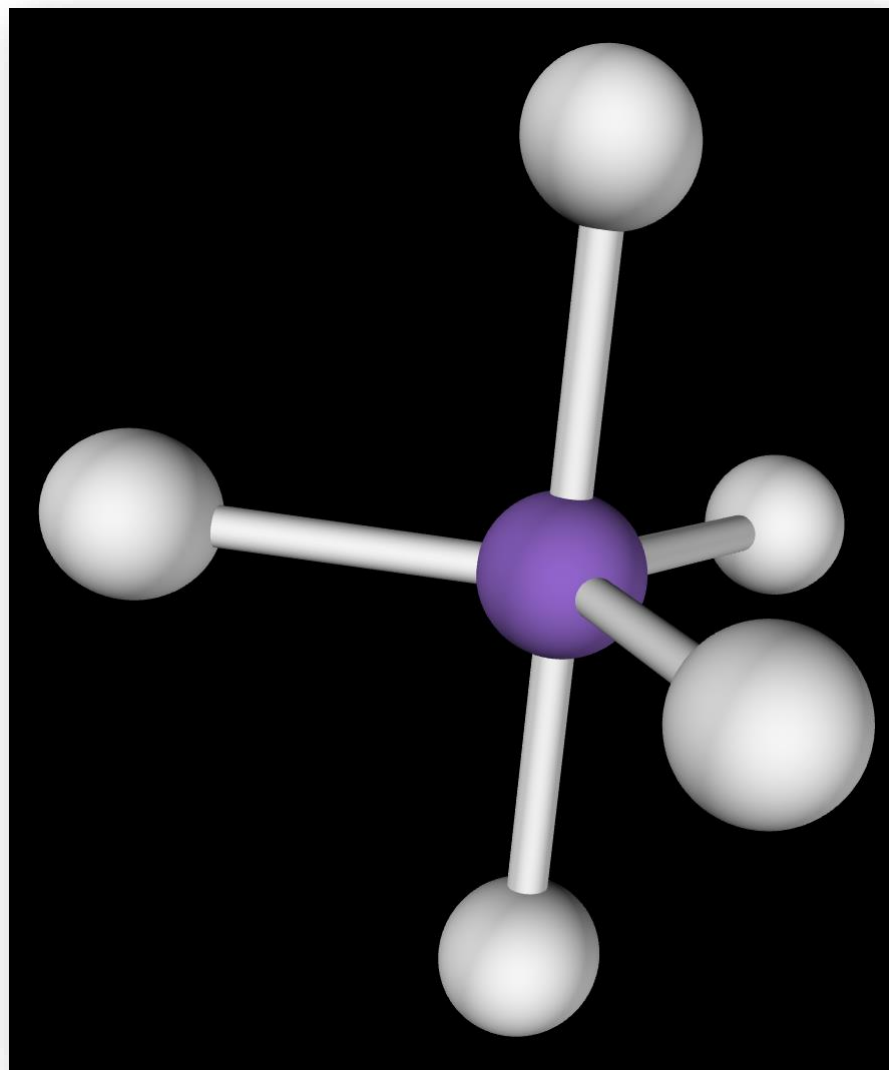
- Electron geometry follows from the **steric number** at the central atom.
https://phet.colorado.edu/sims/html/molecule-shapes/latest/molecule-shapes_all.html
- **Trigonal bipyramidal** electron geometry (steric number 5)



- Important to distinguish between **axial** and **equatorial** positions

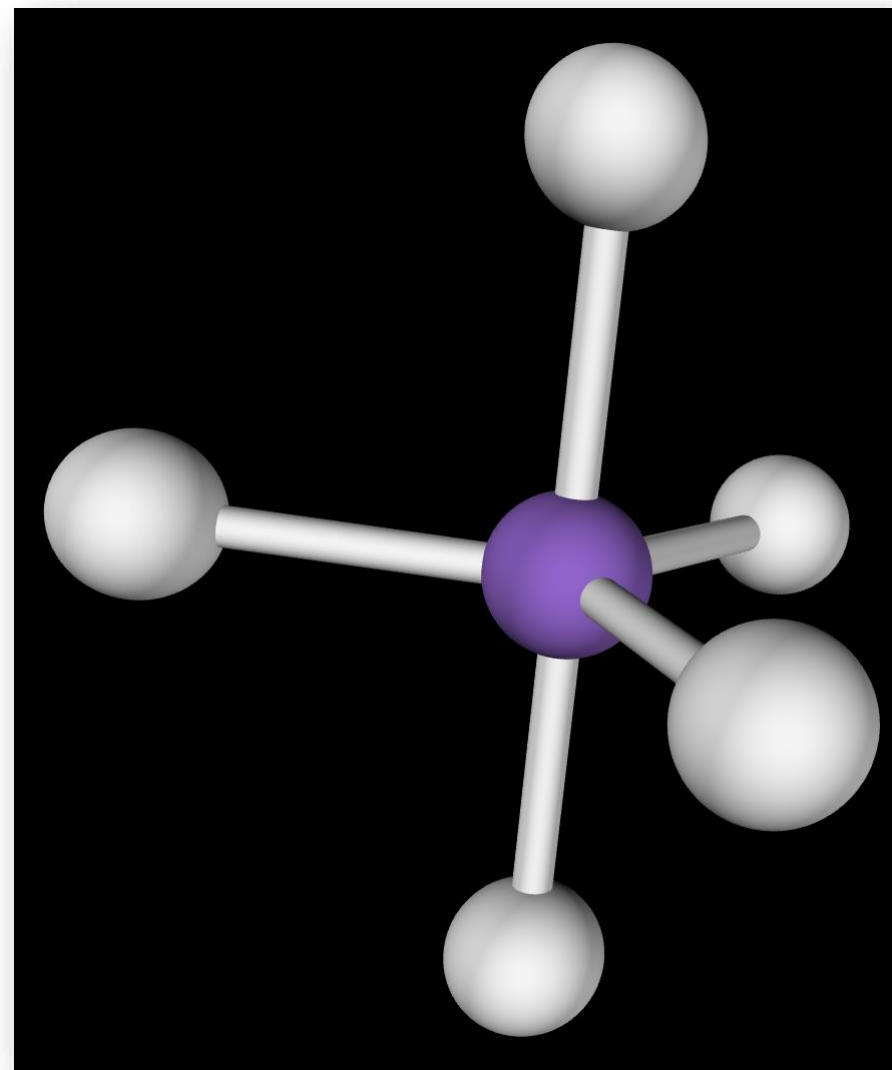
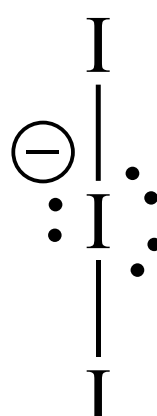
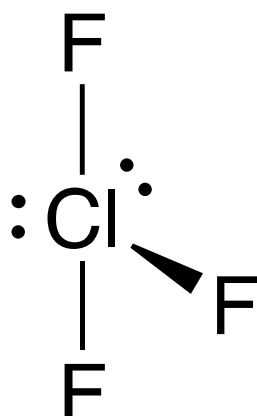
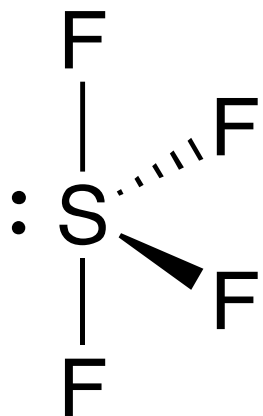


- For trigonal bipyramidal molecules with lone pairs, where do the lone pairs go?
- **Example.** Draw a three-dimensional structure for SF_4 .



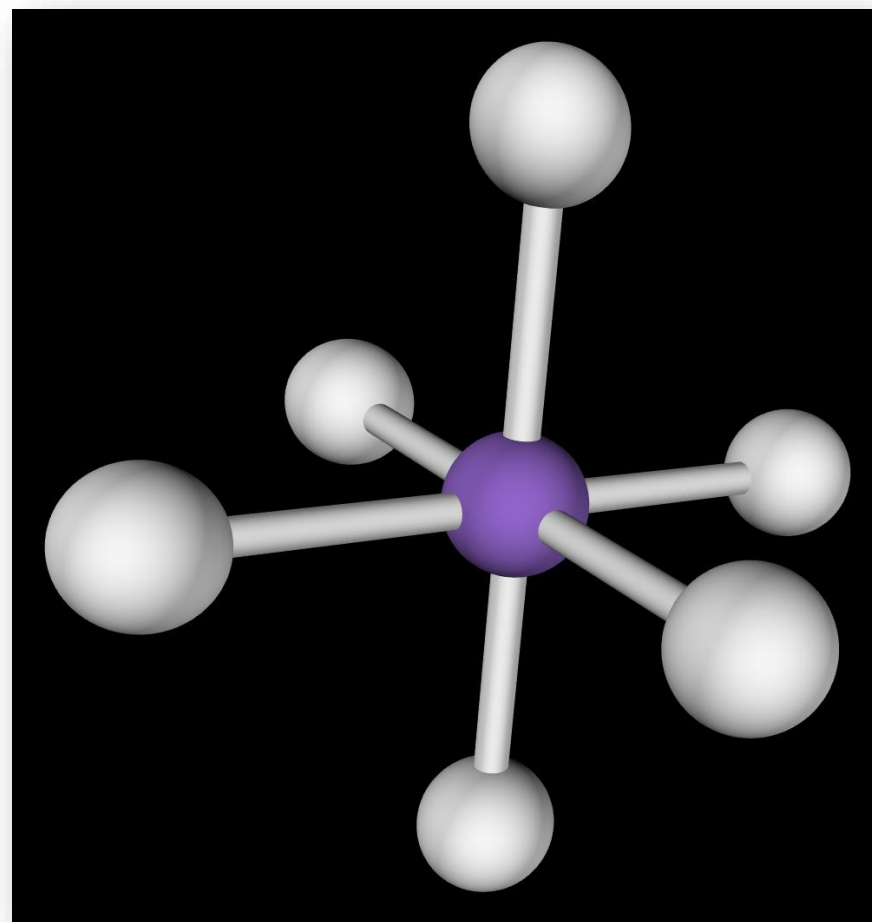
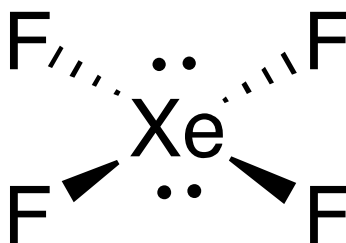
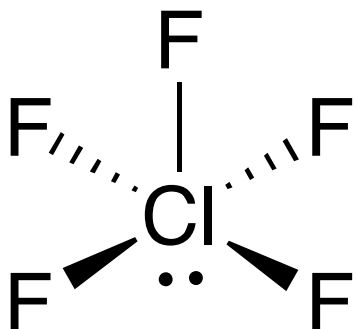
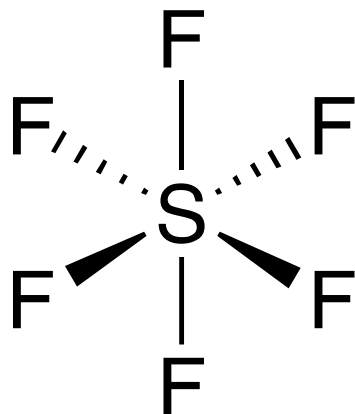
11.1 Lone Pairs in the Trigonal Bipyramidal Geometry | 2 of 2

- Electron geometry follows from the **steric number** at the central atom.
https://phet.colorado.edu/sims/html/molecule-shapes/latest/molecule-shapes_all.html
- **Trigonal bipyramidal** electron geometry (steric number 5)
- Lone pairs occupy the equatorial positions in the most stable structure. This leads to predictable molecular geometries...



11.1 Octahedral Geometry

- Electron geometry follows from the **steric number** at the central atom.
https://phet.colorado.edu/sims/html/molecule-shapes/latest/molecule-shapes_all.html
- **Octahedral** electron geometry (steric number 6)



11.1 Summary of Geometries

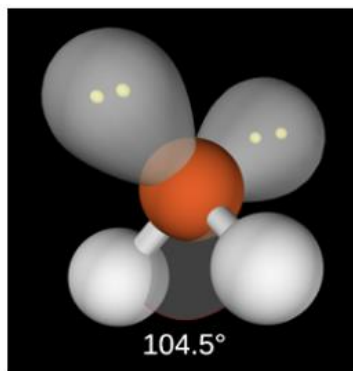
- Note the deviations (“<”) from ideal bond angles—why are these observed?!

TABLE 11.1 Summary of Electron Geometries and Molecular Geometries

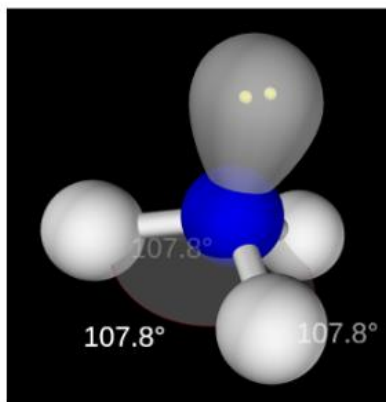
Electron Domains	Bonded Groups	Lone Pairs	Electron Geometry	Molecular Geometry	Bond Angles	Example	Model
2	2	0	Linear	Linear	180°	CO ₂	
3	3	0	Trigonal planar	Trigonal planar	120°	NO ₃ ⁻	
	2	1	Trigonal planar	Bent	< 120°	NO ₂ ⁻	
4	4	0	Tetrahedral	Tetrahedral	109.5°	CH ₄	
	3	1	Tetrahedral	Trigonal pyramidal	< 109.5°	NH ₃	
	2	2	Tetrahedral	Bent	< 109.5°	H ₂ O	
5	5	0	Trigonal bipyramidal	Trigonal bipyramidal	90°, 120°, 180°	PCl ₅	
	4	1	Trigonal bipyramidal	Seesaw	< 90°, < 120°, < 180°	SF ₄	
	3	2	Trigonal bipyramidal	T-shaped	< 90°, < 180°	ClF ₃	
	2	3	Trigonal bipyramidal	Linear	180°	I ₃ ⁻	
6	6	0	Octahedral	Octahedral	90°, 180°	SF ₆	
	5	1	Octahedral	Square pyramidal	< 90°, < 180°	ClF ₅	
	4	2	Octahedral	Square planar	90°, 180°	XeF ₄	

11.1 Deviations from Ideal Bond Angles

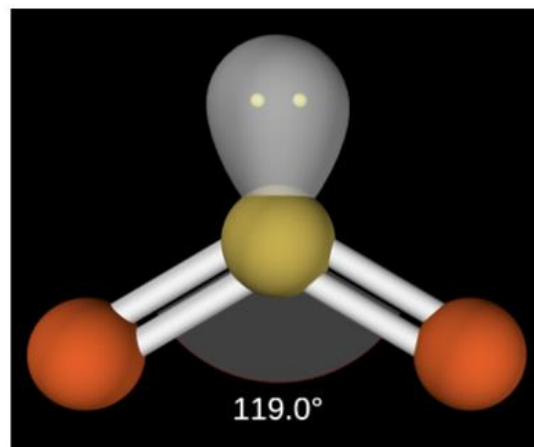
- Lone pairs are larger than bonding pairs. Lone pairs “scrunch” bonding domains closer together, leading to real bond angles slightly less than the ideal angles in molecules containing “asymmetrically positioned” lone pairs.



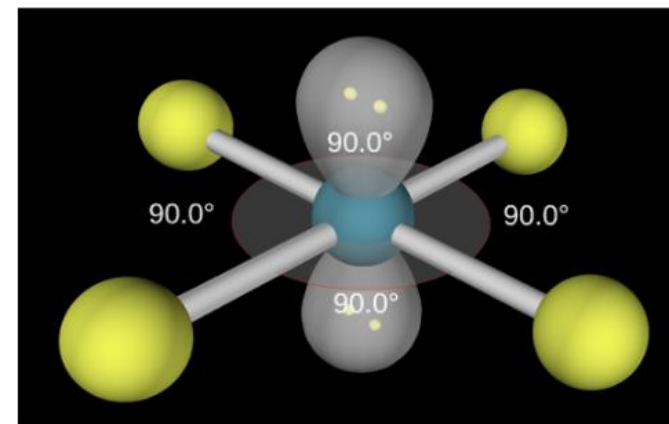
H₂O



NH₃



SO₂



XeF₄

- Note that the effects of symmetrically disposed lone pairs cancel one another!

11.1 Dealing with Multiple Central Atoms

- For a molecule with multiple central atoms, the overall molecular shape is described as the set of shapes around each atom linked to each other.

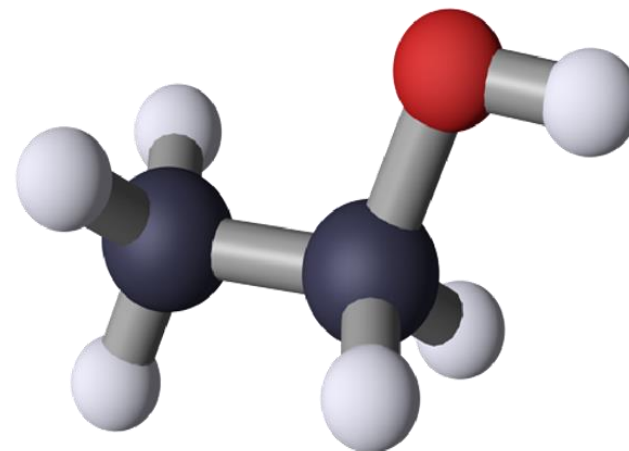
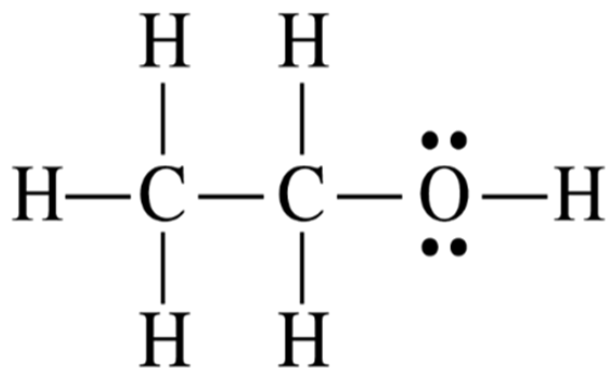


Figure 11.10. Lewis structure and molecular shape of ethanol.

Select all the molecules from the list below that contain at least one 90° bond angle, considering the *ideal* VSEPR geometries.



Consider a molecule with bent molecular geometry and a bond angle of 117.5° . Which statements are *true* regarding this molecule? Select all that apply.

- A. The electron geometry is tetrahedral.
- B. The electron geometry is trigonal planar.
- C. The electron geometry cannot be determined.
- D. The molecule has four *total* electron domains.
- E. The molecule has three *total* electron domains.

Chapter 11.B

Molecular Shape and Bonding Theories

Chemical Principles I

CHEM 1211K

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7. Describe the limitations of valence bond theory.
8. Describe and apply the tenets of molecular orbital theory.
9. Determine bond order and magnetic properties using molecular orbital diagrams.

Why does a molecule with 6 regions of electron density and 2 lone pairs adopt a square planar molecular shape? Select all statements that are *true*.

- A. The total electron-electron repulsion is minimized by placing lone pairs in equatorial positions.
- B. An angle of 180° between lone pairs places them as far away from each other as possible.
- C. The bonding domains adopt bond angles of 90° (rather than smaller bond angles) to minimize the electron-electron repulsion amongst themselves.
- D. The higher repulsion between bonding domains compared to between lone pairs forces the lone pairs to the axial positions.

10.5 Polar Bonds and the Bonding Continuum (Review)

- When two atoms of different electronegativity engage in a covalent bond, one atom has greater electron density than the other.
- One end of the bond has partial positive charge and the other partial negative charge—a **dipole**.
- **Polar covalent bonds** have substantial dipoles.

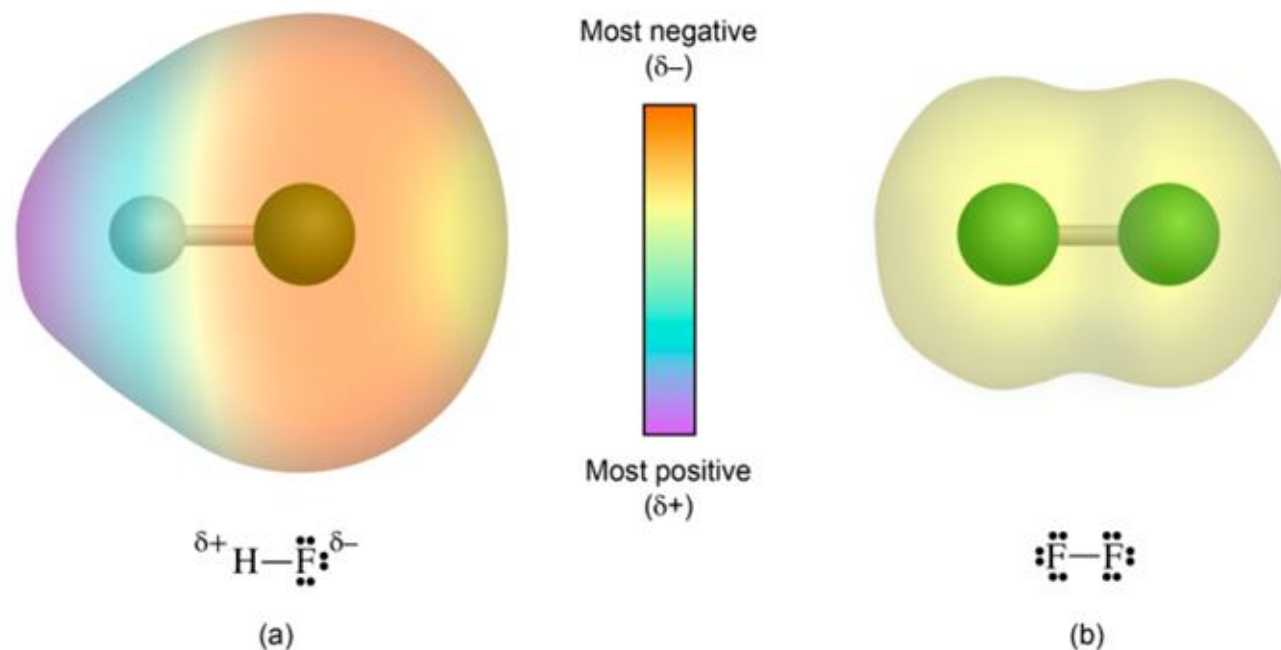


Figure 10.16. (a) Unequal electron sharing in HF and (b) equal electron sharing in F_2 .

11.2 Molecular Dipole Moment; Polar and Nonpolar Molecules



- **Molecular dipole** can be determined by combining knowledge of...
- Molecules with significant dipole moments are **polar**;
molecules with zero or negligible dipole moments are **nonpolar**.

11.2 Polar and Nonpolar Molecules



- Molecules that contain *only* nonpolar bonds are nonpolar.
- Molecules that contain polar bonds may be polar or nonpolar, depending on geometry!
 - A polar molecule has...
 - A nonpolar molecule has...

11.2 Geometry and Polarity

- Molecules with **two or more polar bonds** may be polar or nonpolar, depending on geometry.

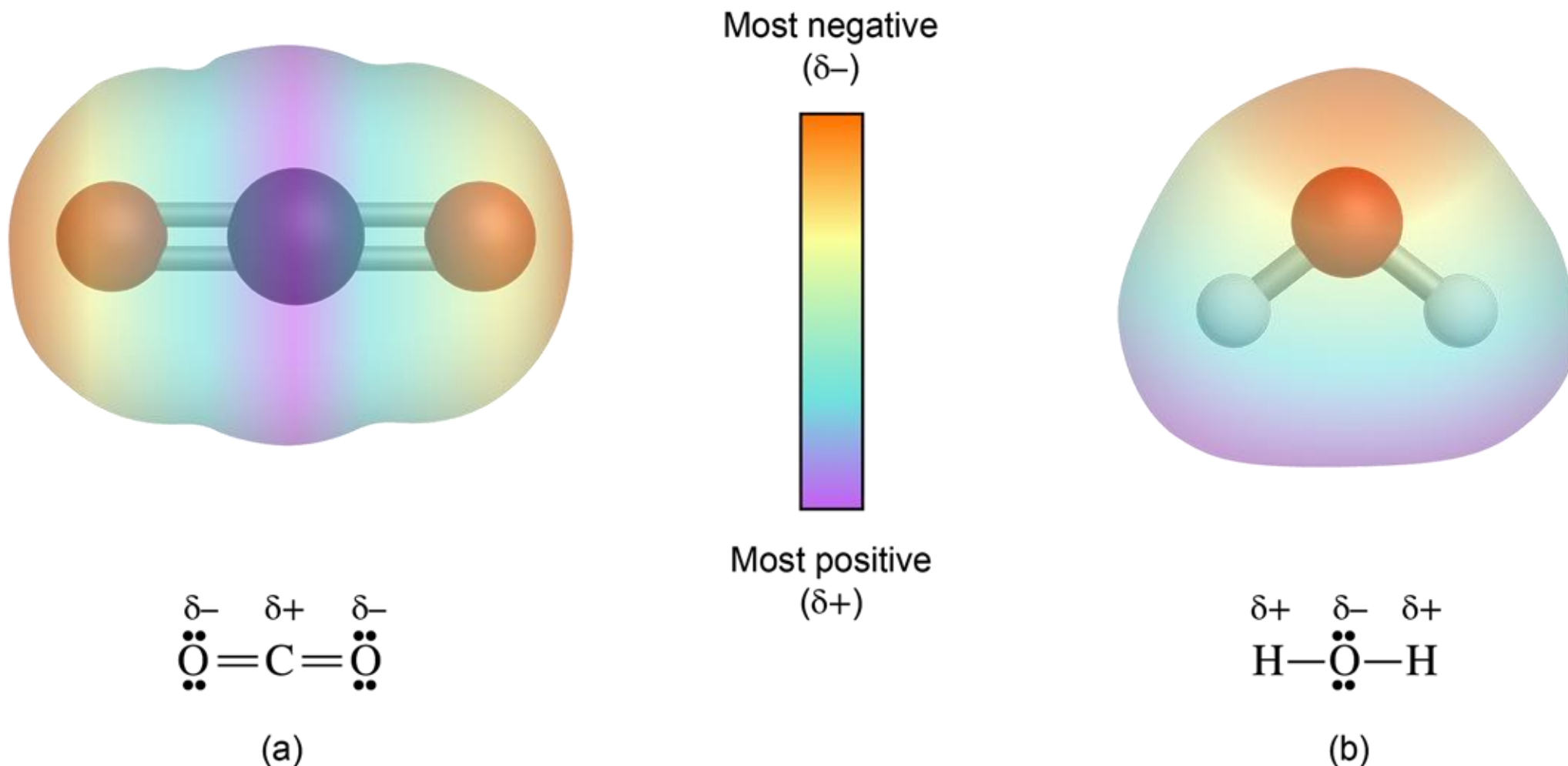


Figure 11.17. Examples of a polar molecule (a) and nonpolar molecule (b) containing polar bonds.

11.2 Practice with Molecular Polarity



Example. Consider the Lewis structure of CH_2Cl_2 . Does this molecule contain polar covalent bonds?
Is CH_2Cl_2 polar or nonpolar?

Consider geometries of the molecules below consistent with VSEPR theory. Which molecules are polar in *all* possible geometries? Enter a string of letters with the necessarily polar molecules in alphabetical order (e.g. "ADEG").



Chapter 11.C

Molecular Shape and Bonding Theories

Chemical Principles I

CHEM 1211K

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8. Describe and apply the tenets of molecular orbital theory.
9. Determine bond order and magnetic properties using molecular orbital diagrams.

Which statement is *true* regarding molecular polarity? For the false statements, generate an example that proves them false.

- A. Molecules with lone pairs of electrons on the central atom must be polar.
- B. Molecules with square planar molecular geometry must be nonpolar.
- C. Tetrahedral molecules with the formula AX_2Y_2 are polar molecules if they contain polar bonds.
- D. Molecules for which the electron geometry and molecular geometry are the same must be nonpolar.
- E. Molecules that contain only nonpolar bonds can be polar or nonpolar.

11.3 Introducing Valence Bond Theory



- **Lewis structures** convey information about connectivity of atoms.
- **VSEPR theory** explains the three-dimensional shape of the molecule.
- **Valence bond theory** provides a model for how covalent bonding takes place—specifically how atomic orbitals and electrons are used to construct covalent bonds.

11.3 Orbital Overlap and Covalent Bonding

- Key aspects of valence bond theory...
 - Orbital overlap
 - Hybrid atomic orbitals
- Covalent bonds are derived from the **overlap** of valence atomic orbitals.
- In diatomic molecules, the atomic orbitals we've already seen overlap to form covalent bonds.

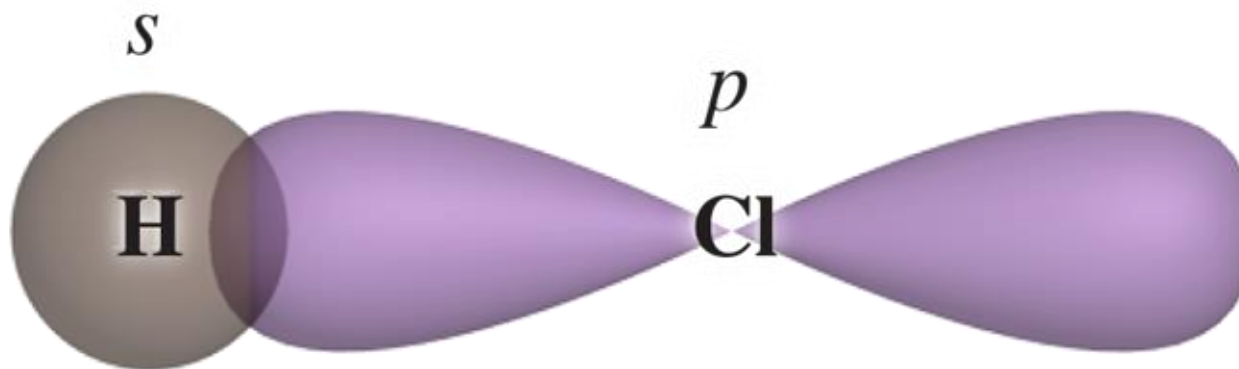


Figure 11.20. Overlap of a 1s and 3p orbital provides a model of the covalent bond in H—Cl.

11.3 Beyond the Atomic Orbitals

- The simple atomic orbitals don't work as well to describe bonding in larger molecules.
- For example, in H_2O we can envision two half-filled $1s$ orbitals on hydrogen overlapping with two half-filled $2p$ orbitals on oxygen.
- What's wrong with this picture?

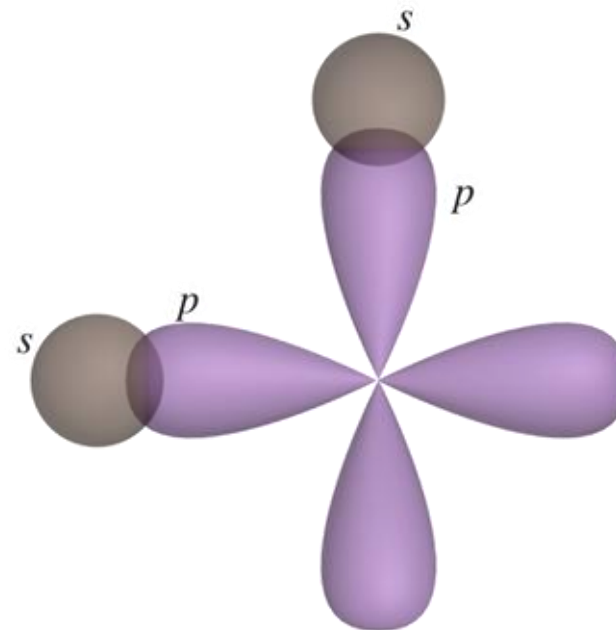
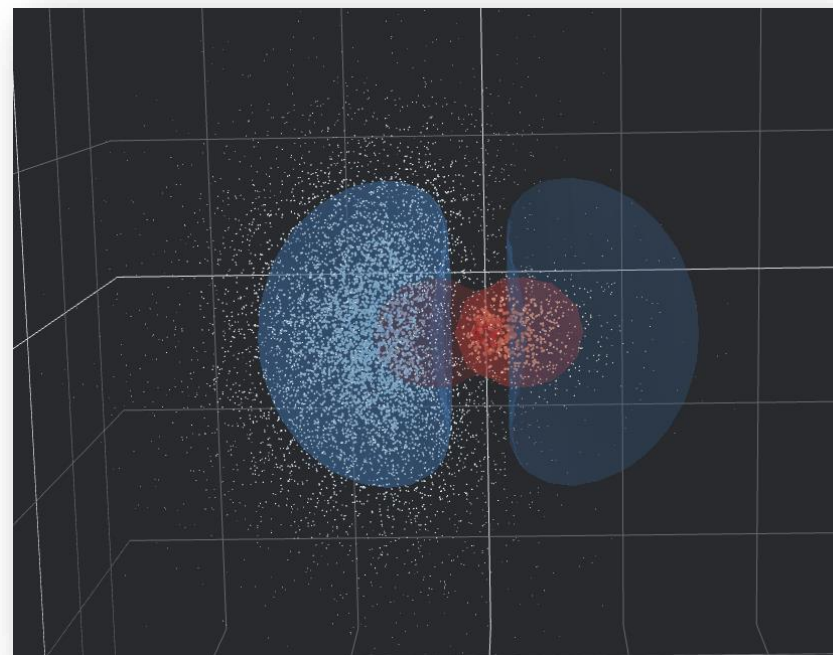


Figure 11.21. Overlap of two $1s$ orbitals with two p orbitals suggests two bonds at a 90° angle.

11.3 Hybrid Atomic Orbitals

- **Hybrid atomic orbitals** are linear combinations of the atomic orbitals on one atom oriented in directions that are consistent with the VSEPR geometries.
 - The number of atomic orbital “inputs” is equal to the number of hybrid orbital “outputs.”
 - Written as sp^n , where 1 s orbital is combined with n p orbitals
 - Used for sigma bonding and nonbonding lone pairs



<https://al2me6.github.io/evanescence/>

11.3 sp^3 Hybridization



- **sp^3 hybridization.** Atoms with four electron domains require four hybrid orbitals, which are constructed from four atomic orbitals ($1 s + 3 p$).
- **Example.** Methane (CH_4)

11.3 Orbital Overlap and Hybrid Atomic Orbitals

- To form covalent bonds, the larger lobe of each hybrid orbital overlaps with another hybrid orbital or atomic orbital.
- Nonbonding lone pairs can occupy hybrid orbitals as well.

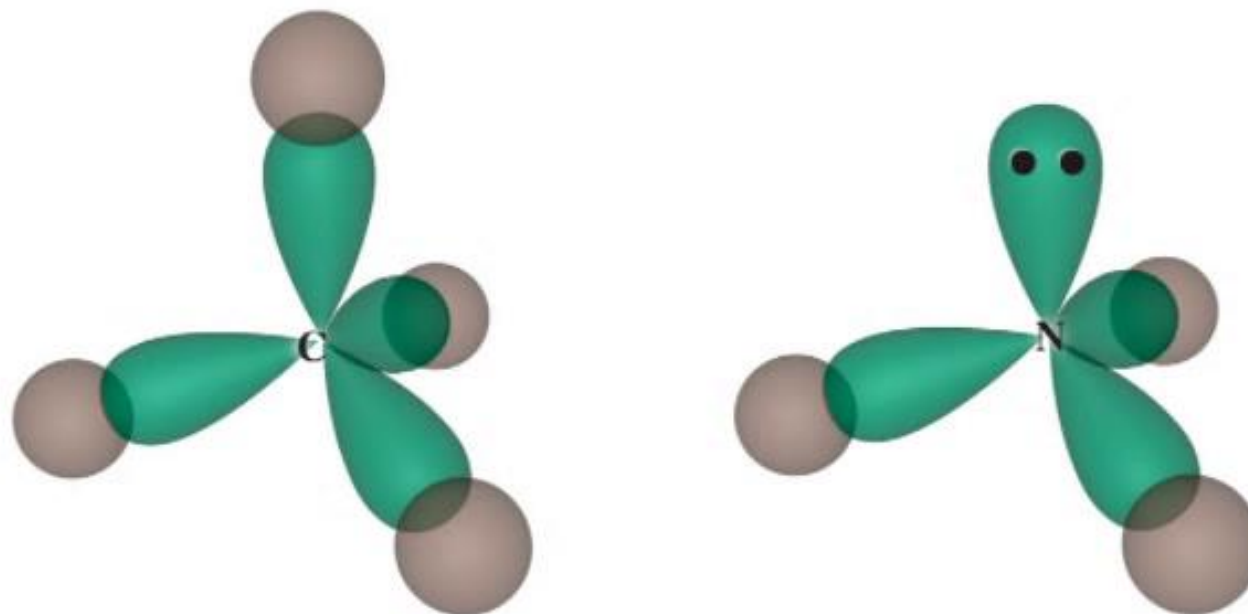


Figure 11.22. Hybrid orbitals (green) overlapping with 1s orbitals on hydrogen in CH₄; a lone pair of electrons occupying a hybrid orbital in NH₃.

11.3 sp^2 Hybridization



- **sp^2 hybridization.** Atoms with three electron domains require three hybrid orbitals, which are constructed from three atomic orbitals ($1s + 2p$).
- **Example.** Borane (BH_3)

- **sp hybridization.** Atoms with two electron domains require two hybrid orbitals, which are constructed from two atomic orbitals ($1s + 1p$).
- **Example.** Hydrogen cyanide (HCN)

11.3 Sigma Bonds

- Bonds constructed from the “head-on” overlap of two hybrid orbitals or atomic orbitals are called **sigma bonds**.
- All covalent bonds involve **one and only one** sigma bond.

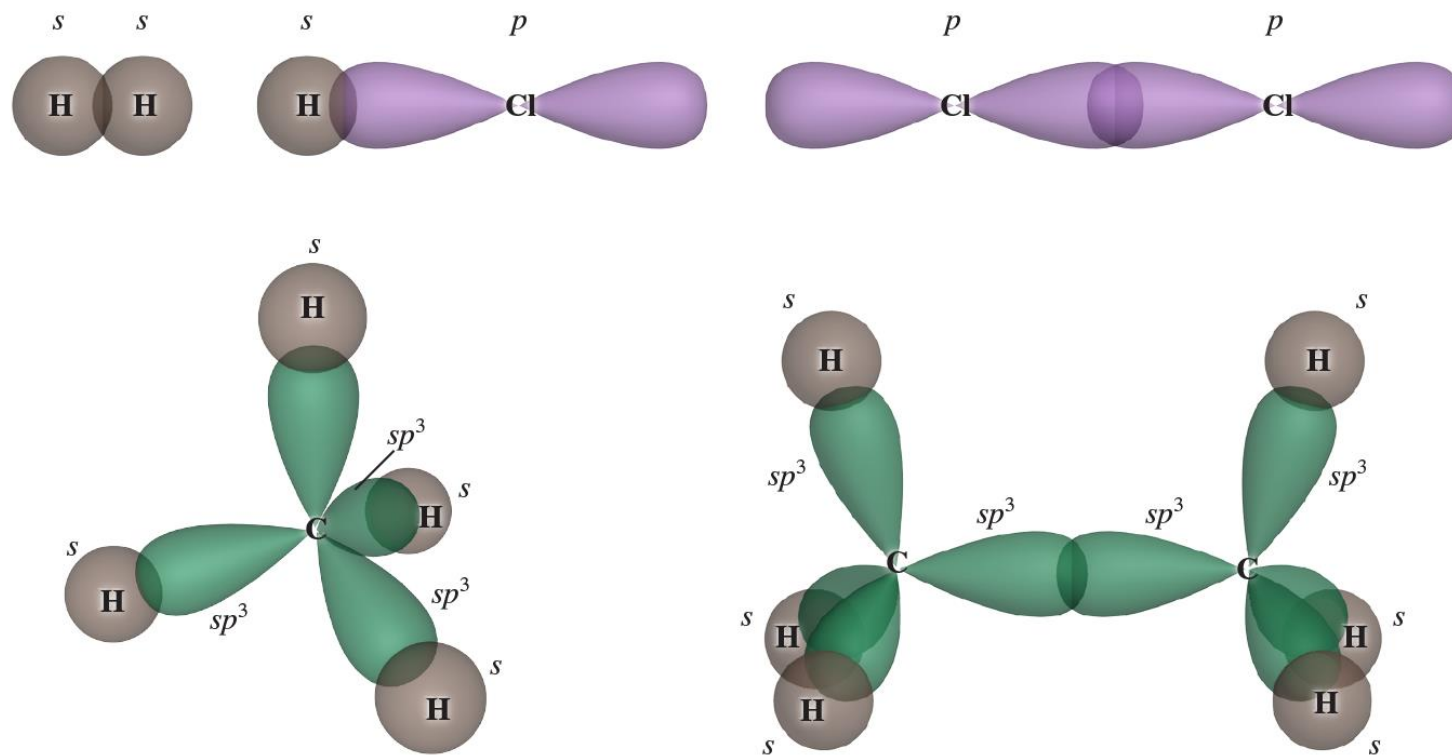
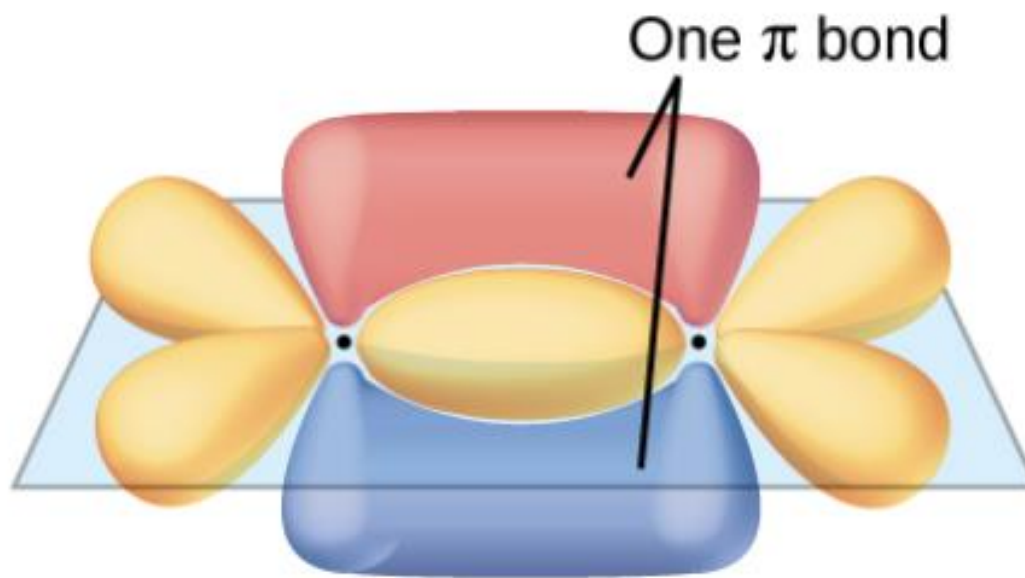


Figure 11.25. Examples of sigma bonds.

11.3 Pi Bonds

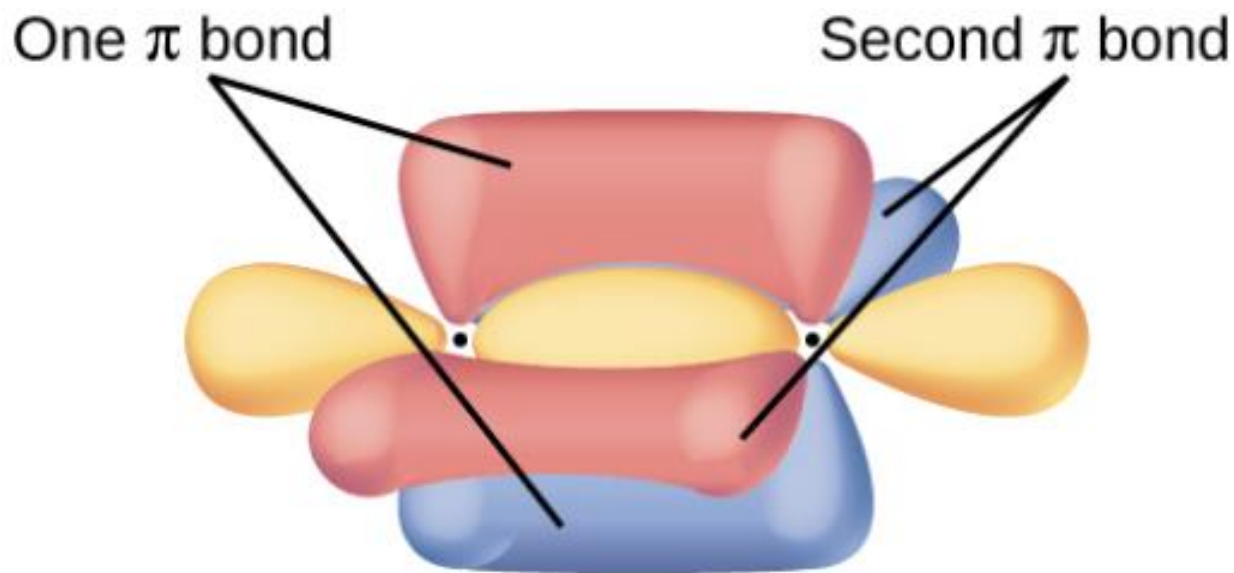
- Unhybridized p orbitals (not used to construct hybrids) overlap in a “side-on” manner to form **pi bonds**.
- Pi bonds are the “second and third bonds” of double and triple bonds.



OpenStax Chemistry 2e, Figure 8.23. Sigma bonds (orange) and pi bond (red and blue) in the ethylene molecule (C_2H_4).

11.3 Pi Bonding in Triple Bonds

- Unhybridized p orbitals (not used to construct hybrids) overlap in a “side-on” manner to form **pi bonds**.
- Two sets of unhybridized p orbitals can overlap to form two distinct pi bonds (in a triple bond).



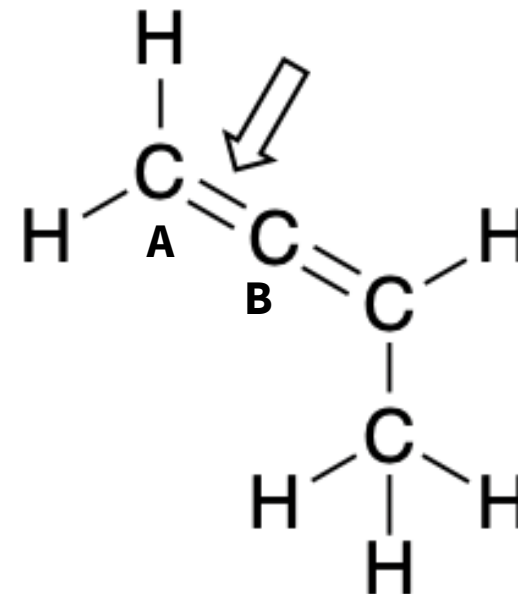
OpenStax Chemistry 2e, Figure 8.25. Sigma bonds (orange) and pi bonds (red and blue) in the ethyne molecule (C₂H₂).

Determine the hybridizations of the carbon atoms involved in the indicated bond. Complete the following sentence and enter the numeral xy as your response.

Carbon **A** has sp^x hybridization and carbon **B** has sp^y hybridization.

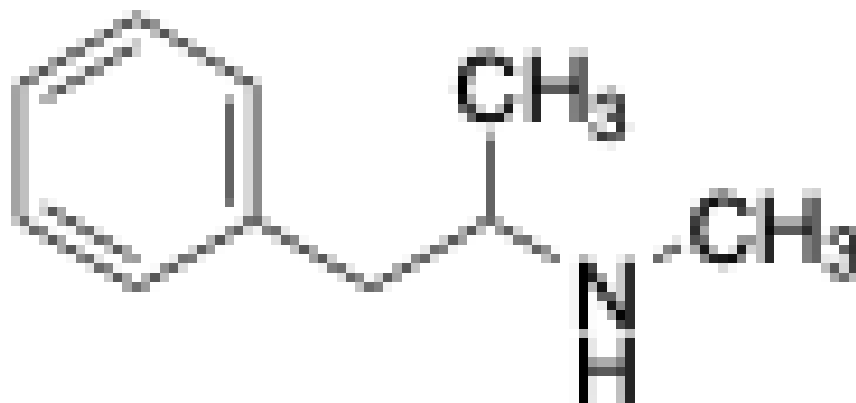
What orbitals overlap to form the sigma bond at the indicated position?

What orbitals overlap to form the pi bond at the indicated position?



In skeletal structures of organic compounds, carbons are located at the intersections of bonds and C—H bonds are omitted under the assumption that neutral carbons satisfy the octet rule. Formal charges are never omitted.

Add missing C—H bonds to the structure of methamphetamine below and determine the hybridization of each atom. How many sp^3 -hybridized atoms does this molecule contain?



11.3 Applications of Valence Bond Theory



- Hybridization does not adequately describe bonding at atoms with more than four electron domains.
Hypervalency is a more accurate model than sp^3d^n hybridization, but we will not cover it in CHEM 1211K.
- Valence bond theory can be used to explain certain observations of chemical structure and reactivity...

11.3 Multiple Bonds are Reactive



- Molecules with multiple bonds tend to be more reactive than those with only single bonds...

11.3 Restricted Rotation of Double Bonds

- Single bonds rotate freely but double bonds do not...

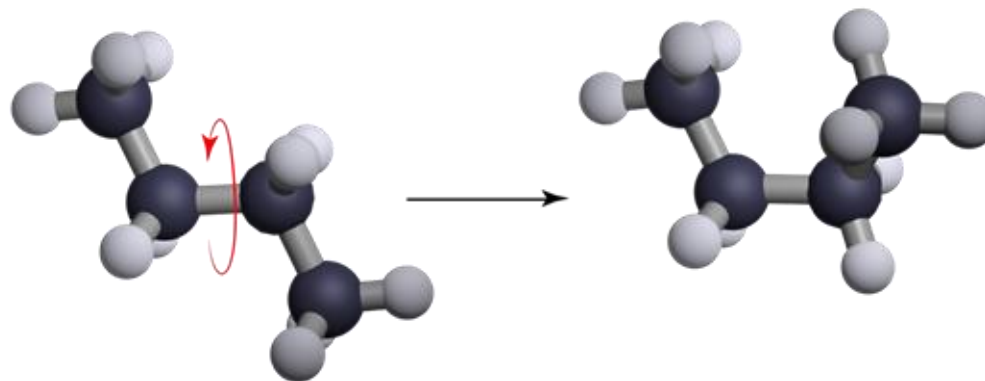


Figure 11.32. The central single bond in butane rotates very rapidly.

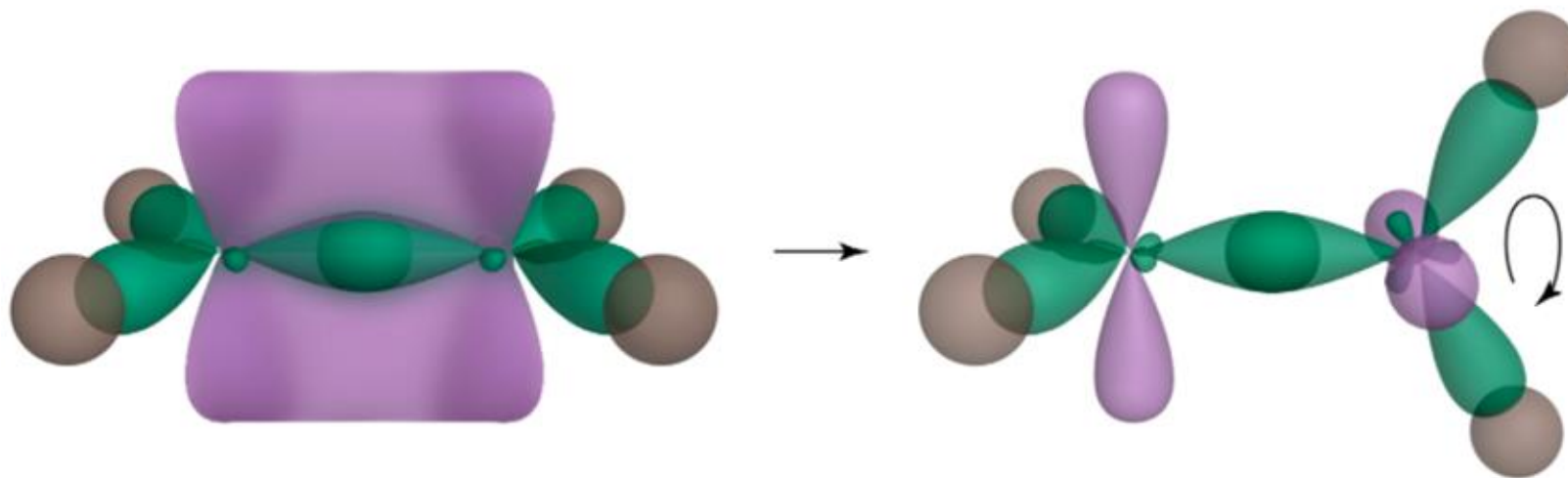


Figure 11.33. Rotation about the double bond in ethylene breaks the pi bond by eliminating orbital overlap.

Chapter 11.D

Molecular Shape and Bonding Theories

Chemical Principles I

CHEM 1211K

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- Valence bond theory does not account for several important observations of molecules, such as...
 - Energy levels of electrons in molecules
 - Magnetic properties
- **Molecular orbital theory** is a bonding theory that explains these properties (and so much more!).

- Your textbook provides a solid overview of molecular orbital theory and additional information can be found in chapter 8 of [OpenStax Chemistry 2e](#).
- Several images in the following lecture slides are from this resource:

Flowers, P., Theopold, K., Langley, R., & Robinson, W. (2019). Advanced Theories of Covalent Bonding. In *Chemistry 2e*. OpenStax: Houston, TX. Retrieved from <https://openstax.org/books/chemistry-2e/pages/preface>.

- Atomic orbitals on different atoms overlap to form **molecular orbitals** that are **delocalized** over the entire molecule.

- Important concepts related to molecular orbitals...

11.5 Constructive and Destructive Orbital Overlap

- Atomic orbitals may overlap in an additive way (**constructive**) or subtractive way (**destructive**), analogous to the way two-dimensional waves may interfere constructively or destructively.

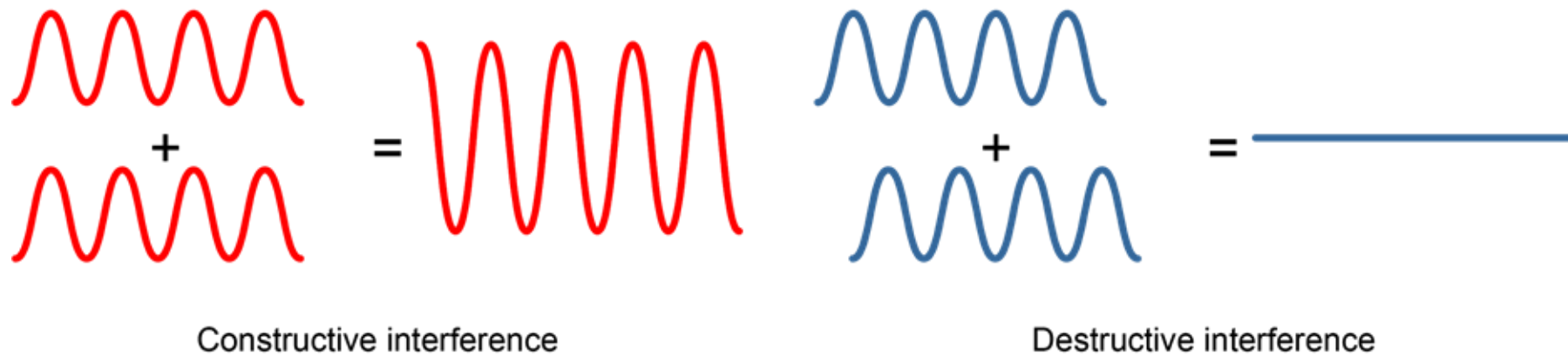
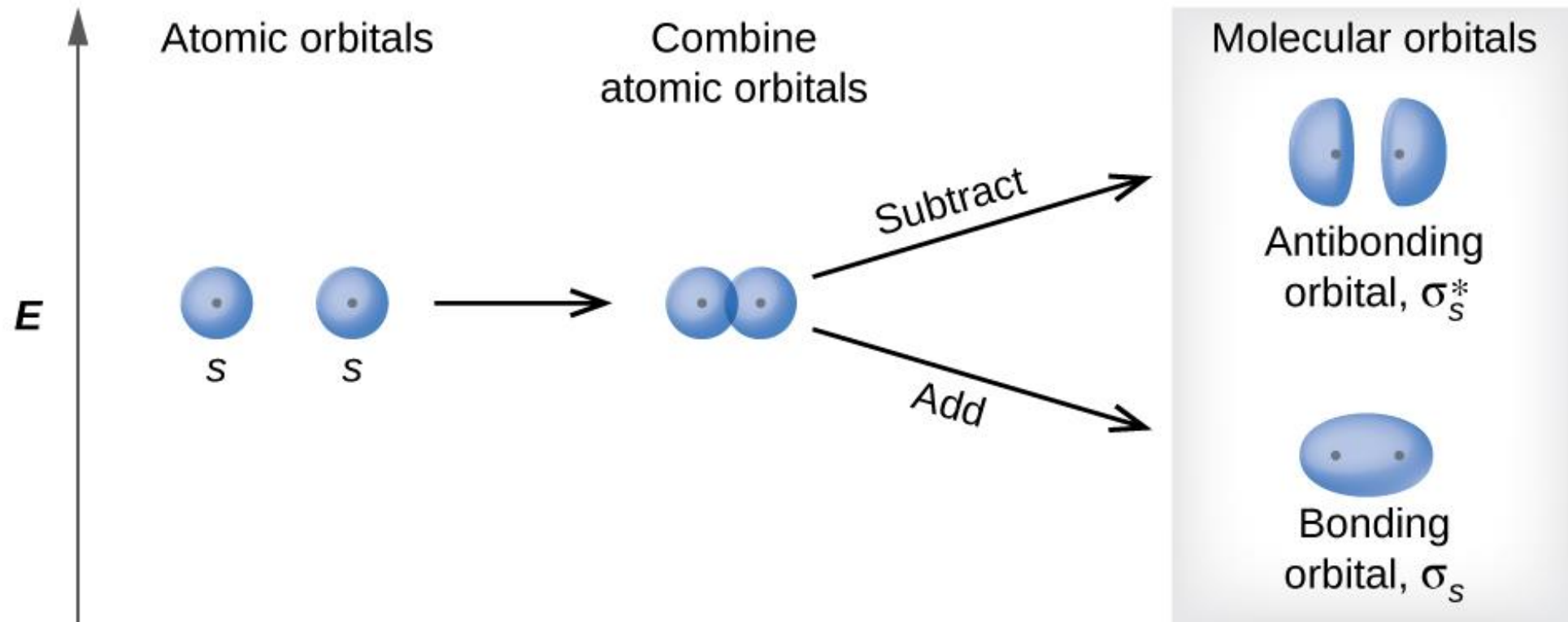


Figure 11.34. Constructive and destructive interference of waves.

11.5 Bonding and Antibonding Molecular Orbitals

- Subtractive overlap (destructive interference) of atomic orbitals results in an **antibonding molecular orbital** that is *higher* in energy than the “input” atomic orbitals.
- Additive overlap (constructive interference) of atomic orbitals results in a **bonding molecular orbital** that is *lower* in energy than the “input” atomic orbitals.



OpenStax Chemistry 2e, Figure 8.29. Combination (overlap) of 1s orbitals on hydrogen results in antibonding and bonding molecular orbitals.

11.5 Molecular Orbital Diagrams; Sigma-type Orbitals

- **Molecular orbital (MO) diagrams** show how atomic orbitals (AOs) are combined to form MOs as well as the relative energies of the AOs and MOs and their occupancies.
- MOs formed from “head-on” overlap of AOs are called **sigma orbitals**.

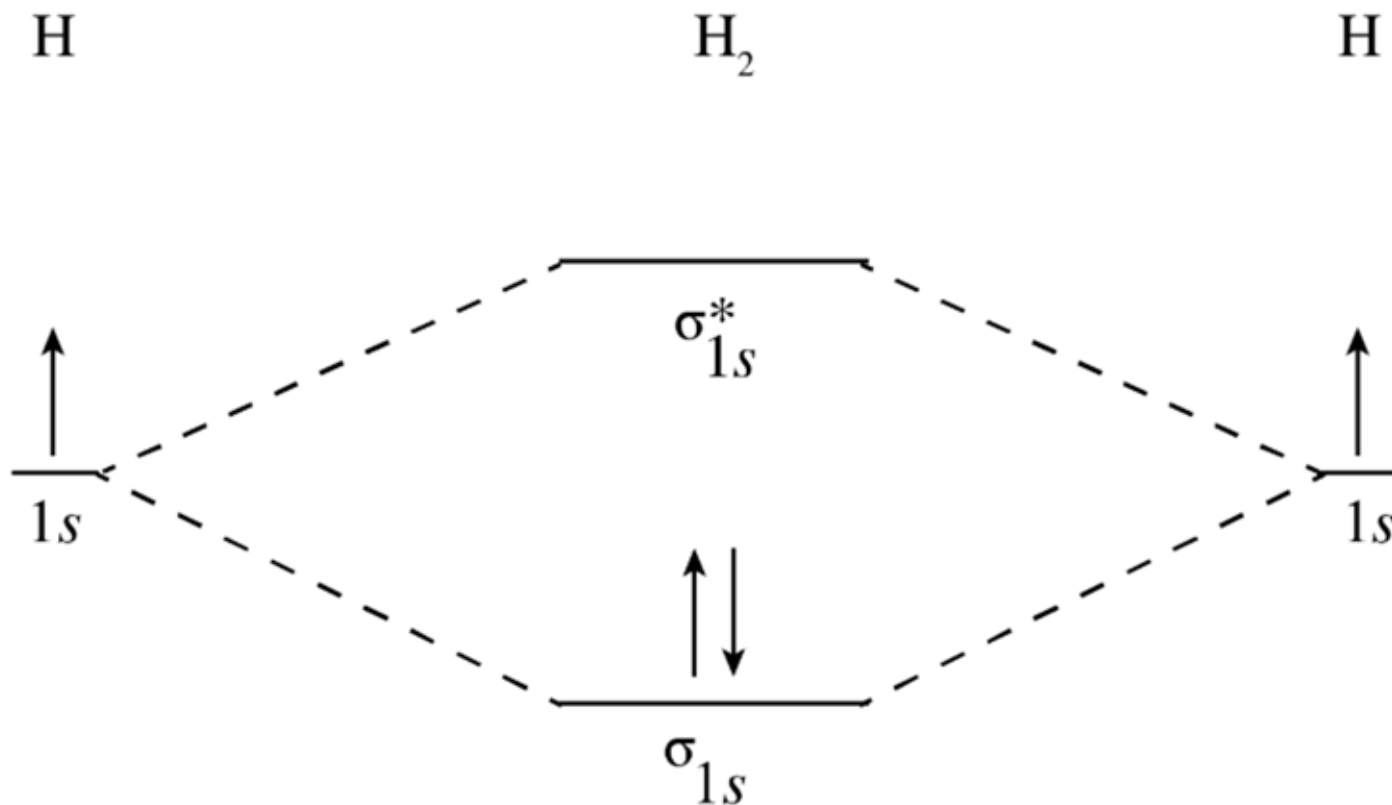


Figure 11.36. Molecular orbital diagram for dihydrogen (H_2).

11.5 Molecular Orbital Electron Configurations

- We can write molecular electron configurations by analogy to atomic electron configurations...

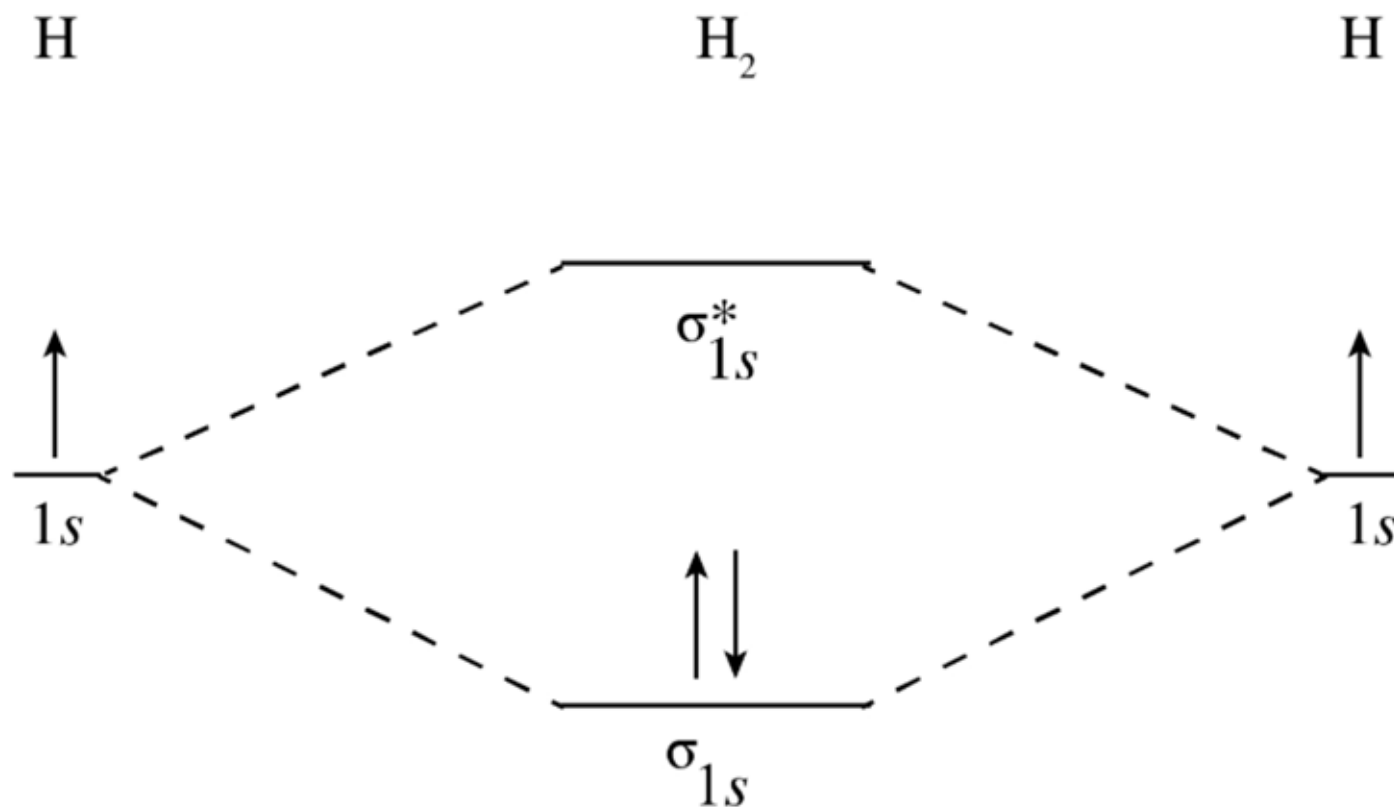
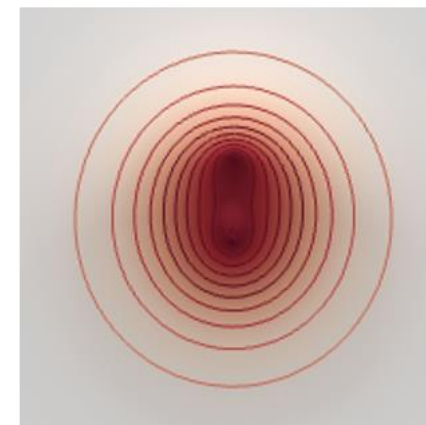


Figure 11.36. Molecular orbital diagram for dihydrogen (H₂).

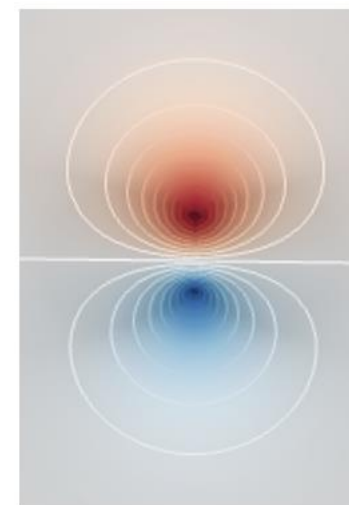
Two-dimensional cross-sections of the molecular orbitals of H_2 are shown at right. Select all the *true* statements from the choices below.

Hint: nuclei are located at the “densest” points in these orbital shapes. Color indicates the sign of the wave function.

- A. Orbital 1 is a σ orbital and orbital 2 is a σ^* orbital.
- B. Both orbitals are derived from the overlap of $2p$ AOs on the hydrogen atoms.
- C. In the neutral H_2 molecule, orbital 1 is empty and orbital 2 contains two electrons.
- D. Orbital 2 is higher in energy than the AOs of which it is composed.
- E. Orbital 2 contains a node between the nuclei.



Orbital 1



Orbital 2

- **Bond order** is a measure of the “number of bonds” between two centers.
- For a diatomic molecule, bond order is equal to half the difference between the numbers of bonding and antibonding electrons.
- Diatomic molecules with a bond order of zero are unstable relative to the separated atoms (cf. He_2).

Draw a molecular orbital diagram for He_2^+ . What is the bond order of this molecule?
Enter your response as a numeral.

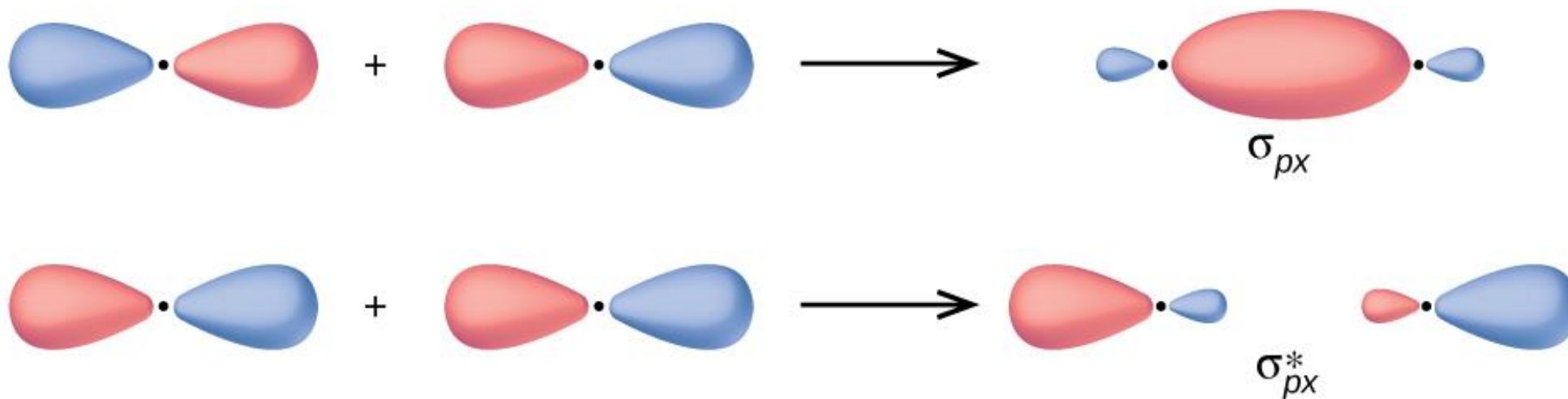
11.5 Molecular Orbitals of Second-period Diatomic Molecules



- So far we have focused on 1s orbitals and the period 1 elements. When the period 2 elements form diatomic molecules, the 2s and 2p AOs combine to form eight **valence MOs**.
- The 2p orbitals overlap to form one pair of sigma orbitals (σ_{2px} , σ_{2px}^*) and two pairs of **pi orbitals** (π_{2py} , π_{2py}^* and π_{2pz} , π_{2pz}^*).
- The 2s orbitals overlap to form a pair of sigma orbitals (σ_{2s} , σ_{2s}^*).

11.5 Sigma-type Orbitals Derived from 2p Orbitals

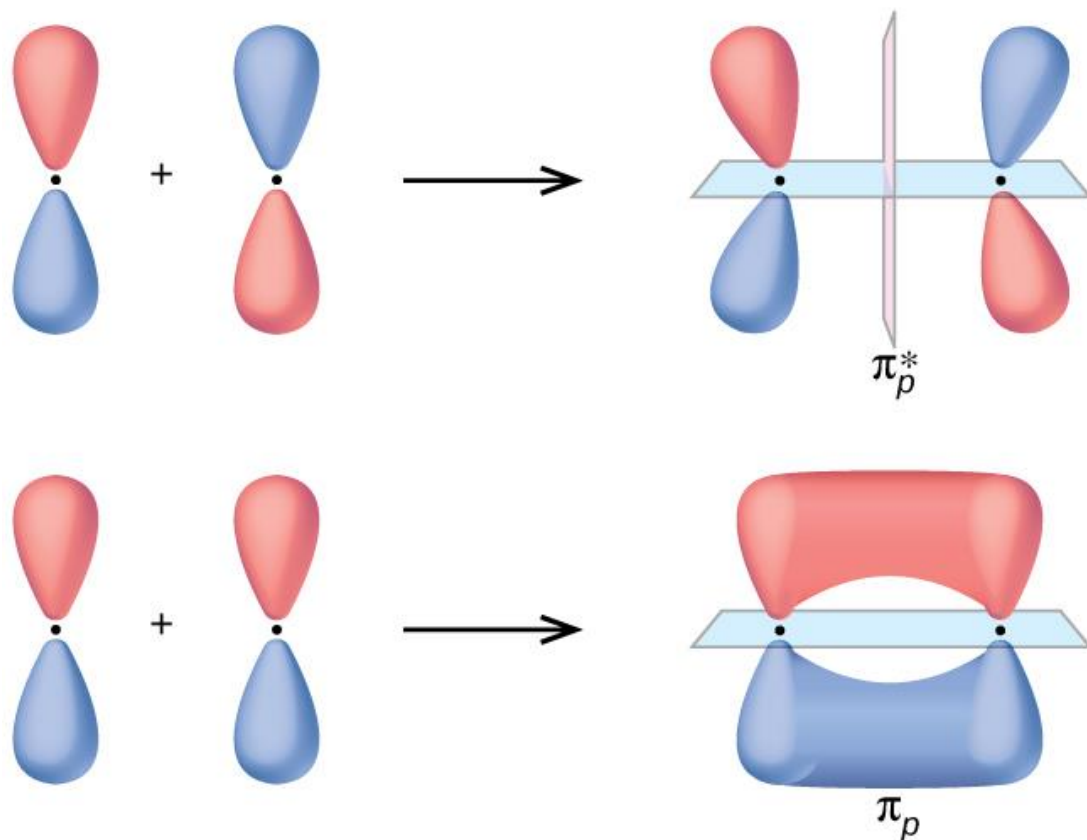
- 2p orbitals that overlap “head on” along the internuclear axis form one pair of sigma orbitals (σ_{2px} , σ_{2px}^*).



OpenStax Chemistry 2e, Figure 8.30. Head-on overlap of $2p_x$ orbitals gives rise to sigma bonding and antibonding MOs.

11.5 Pi-type Orbitals Derived from 2p Orbitals

- 2p orbitals that overlap “side on” form two pairs of pi orbitals (π_{2py} , π^*_{2py} and π_{2pz} , π^*_{2pz}).



OpenStax Chemistry 2e, Figure 8.31. Side-on overlap of 2p orbitals gives rise to pi bonding and antibonding MOs.

Chapter 11.E

Molecular Shape and Bonding Theories

Chemical Principles I

CHEM 1211K

1. Use Lewis structures to predict molecular geometry using VSEPR.
2. Describe and apply VSEPR geometries and deviations from them.
3. Identify polar and nonpolar molecules using electronegativity values and molecular shape.
4. Explain valence bond theory.
5. Describe and recognize sp , sp^2 , and sp^3 hybridization based on the Lewis structure of a molecule.
6. Define, describe, and identify sigma and pi bonds in molecules.
7. Describe the limitations of valence bond theory.
8. Describe and apply the tenets of molecular orbital theory.
9. Determine bond order and magnetic properties using molecular orbital diagrams.

- Molecular orbital diagrams for diatomic molecules of the period 2 elements (B_2 through Ne_2)

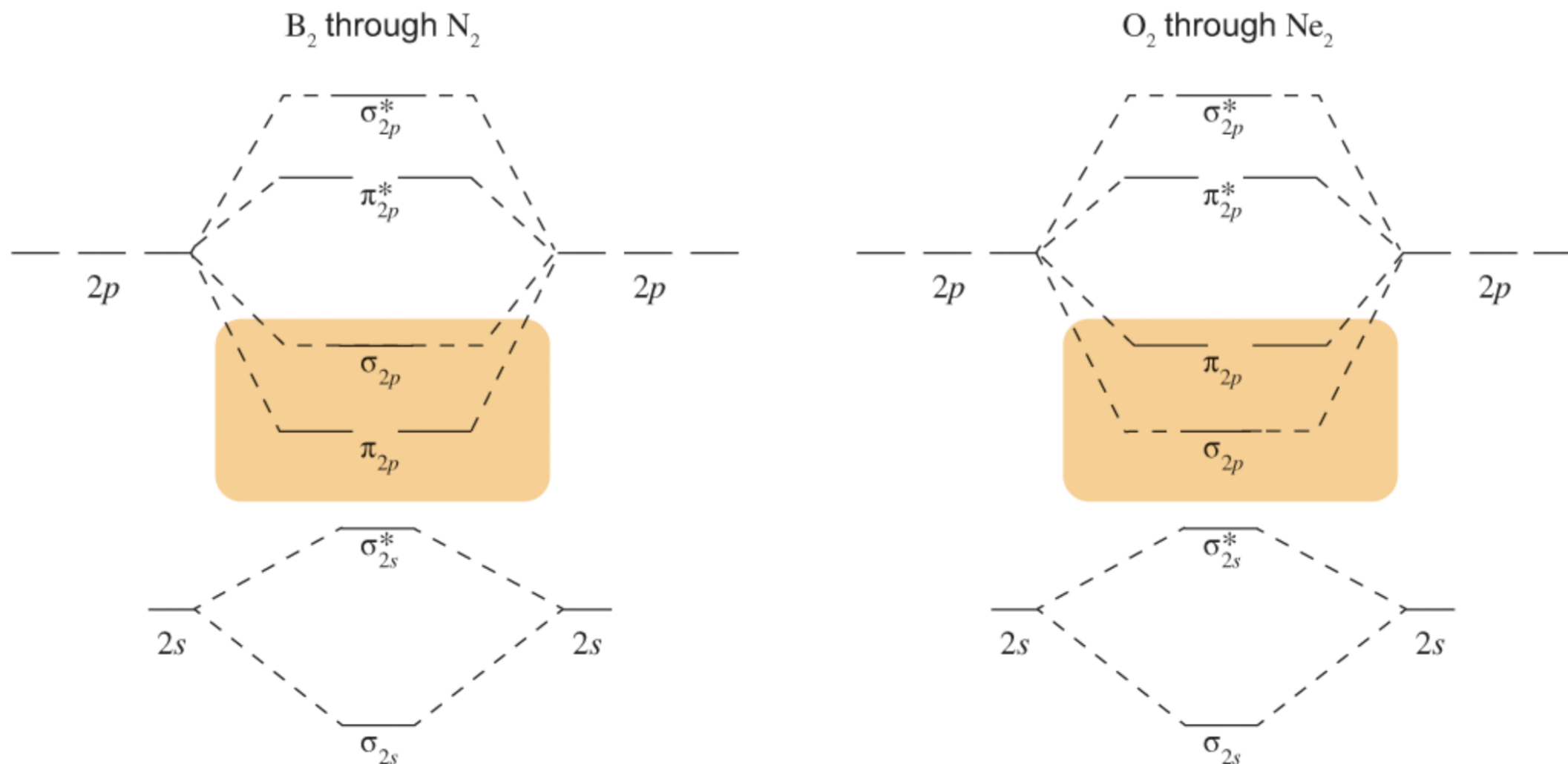
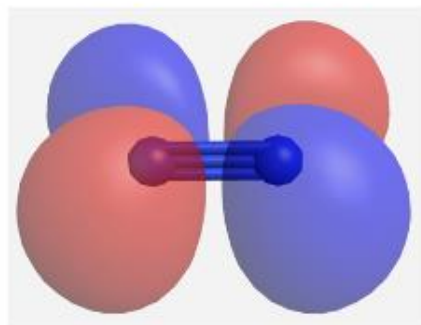
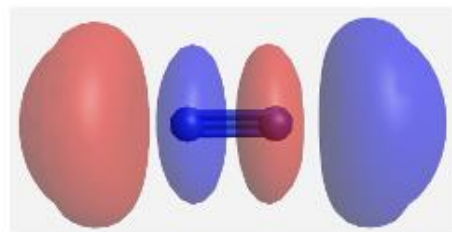
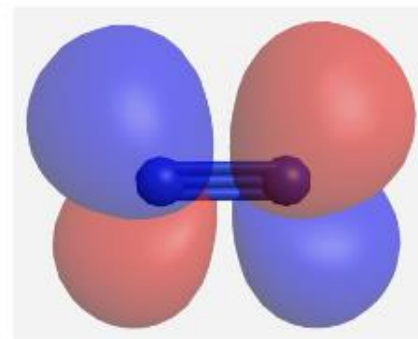
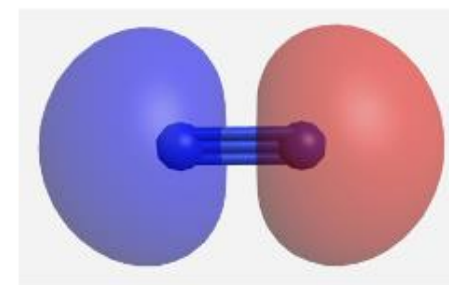
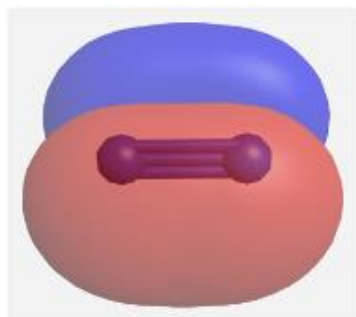
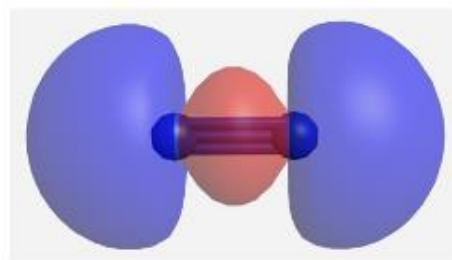
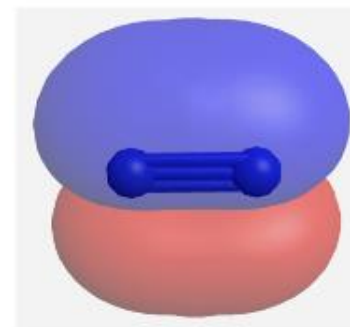
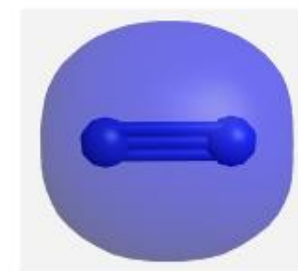


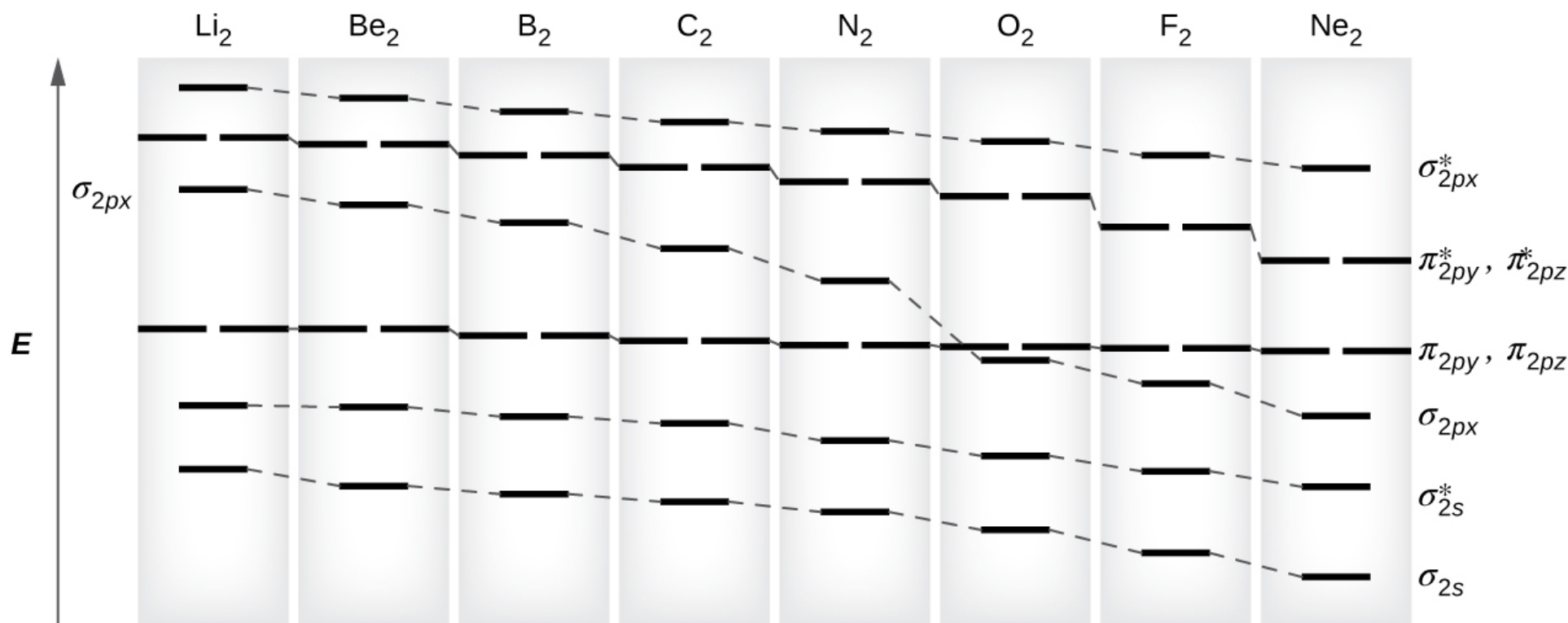
Figure 11.41. Haim, A. J. *Chem. Educ.* **1991**, 68, 737. ([link](#))

- Shapes of the molecular orbitals of the homonuclear diatomics.
Each label corresponds to the indicated shape regardless of the atoms involved.


 π^*_{2py}

 σ^*_{2p}

 π^*_{2pz}

 σ^*_{2s}

 π_{2py}

 σ_{2p}

 π_{2pz}

 σ_{2s}

11.5 Electronegativity and s - p Mixing

- Periodic trends account for the observed “inversion” in the π_{2p} and σ_{2p} energies...
 - Atoms of relatively *low* electronegativity are affected by [s- \$p_x\$ mixing](#), which pushes up the energy of the σ_{2p_x} MO.
 - The energetic ordering observed for O_2 , F_2 , and Ne_2 is “standard.”
 $\sigma \rightarrow \pi \rightarrow \pi^* \rightarrow \sigma^*$ is the ordering that will become default in organic chemistry.



OpenStax Chemistry 2e, Figure 8.37. Relative energies of orbitals in the second-row diatomic molecules.

Example. Determine the MO electron configuration and bond order of N_2 .
Does the bond order agree with the classical Lewis structure?

Example. Determine the MO electron configuration and bond order of O_2 .
Does the bond order agree with the classical Lewis structure?

- Removing or adding electrons (oxidation or reduction) can affect bond order.
 - Bond order increases when bonding electrons are added or antibonding electrons removed.
 - Bond order decreases when antibonding electrons are added or bonding electrons removed.
- Excitation of a molecule by light also causes a change in bond order when the occupancies of bonding and antibonding orbitals change.

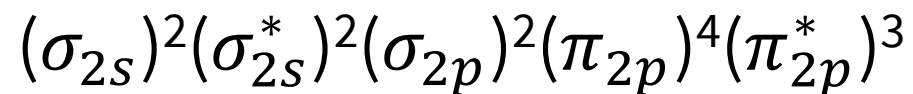
11.5 Practice with Molecular Orbital Theory



Example. Which molecule contains a stronger bond, O_2 or O_2^{2+} ? Which molecule contains a longer bond?

How many electrons does N_2^{2+} have in valence antibonding orbitals?

Which molecule or ion could have the ground state molecular orbital electron configuration below? Assume that orbitals are listed from lowest to highest energy.



A. F_2

B. N_2^{3-}

C. F_2^+

D. N_2

E. O_2^{2-}

Chapter 11.F

Molecular Shape and Bonding Theories

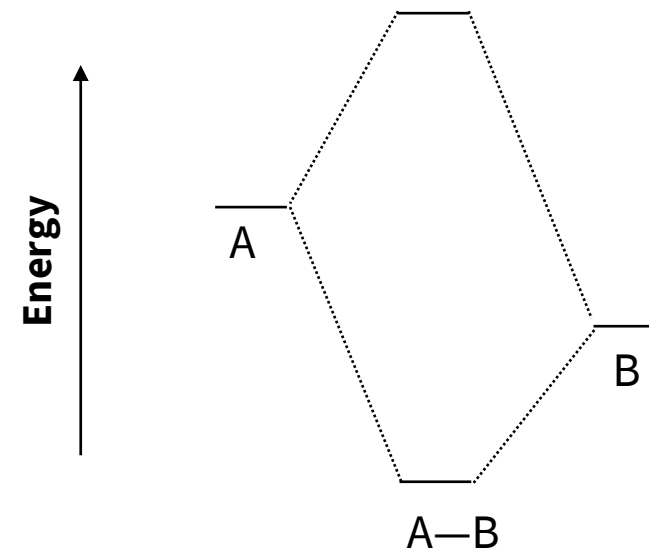
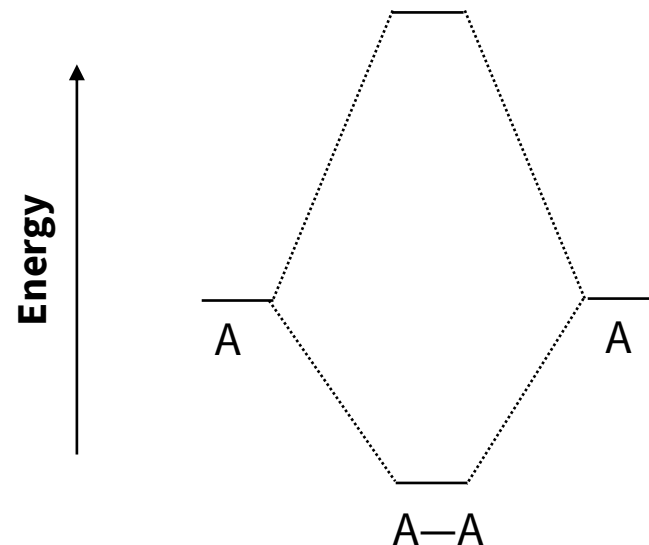
Chemical Principles I

CHEM 1211K

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6. Define, describe, and identify sigma and pi bonds in molecules.
7. Describe the limitations of valence bond theory.
8. Describe and apply the tenets of molecular orbital theory.
9. Determine bond order and magnetic properties using molecular orbital diagrams.

11.5 Heteronuclear Diatomic Molecules

- In a **homonuclear** diatomic molecule, the energies of overlapping AOs are equal and “MO splitting” is large. MOs are symmetrical in shape.
- In a **heteronuclear** diatomic molecule, the energies of overlapping AOs are unequal. MO splitting is relatively small and each MO has greater contribution from one atom or the other.



11.5 Nitrogen Monoxide and Hydrogen Fluoride

Example. Molecular orbitals of nitrogen monoxide and hydrogen fluoride

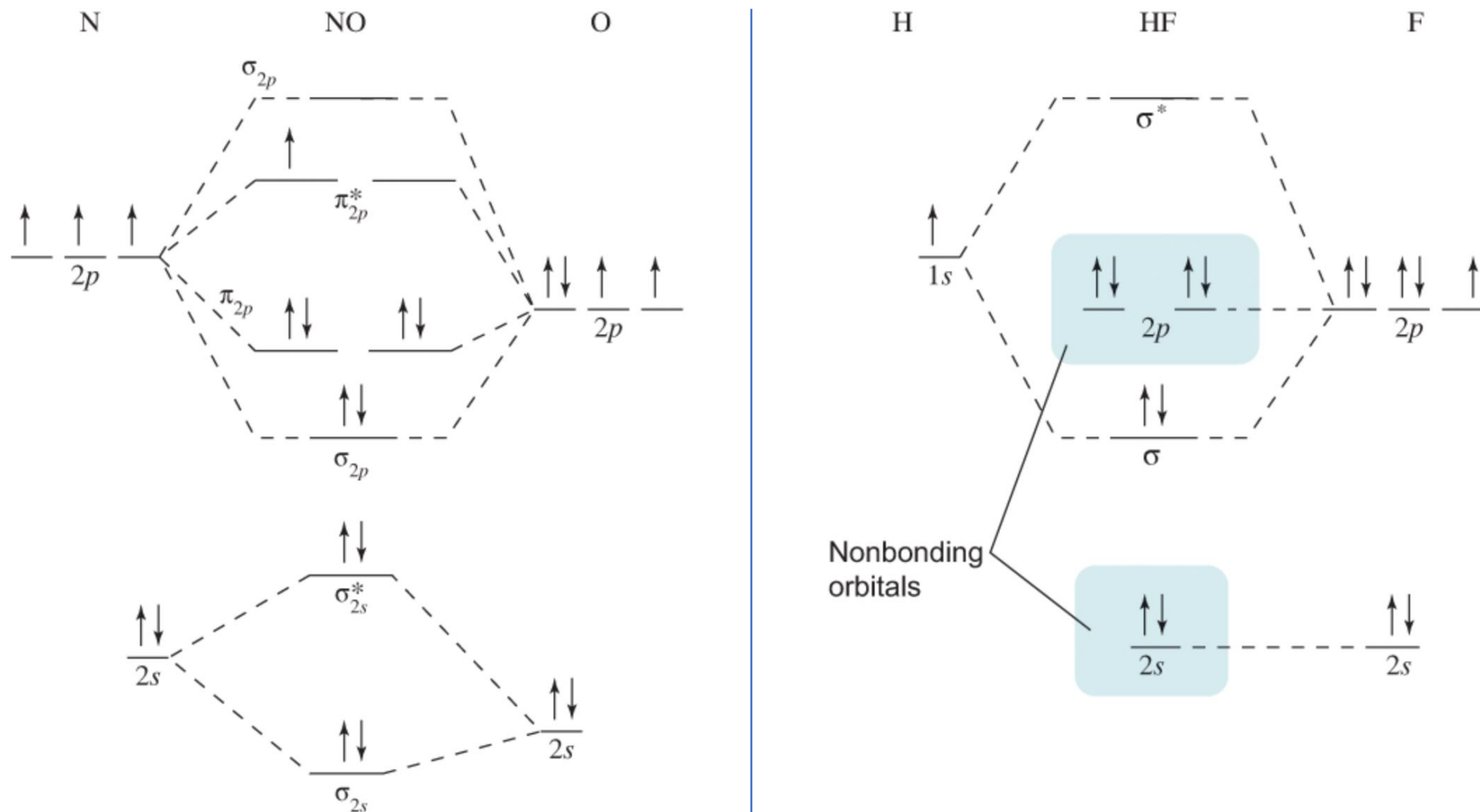
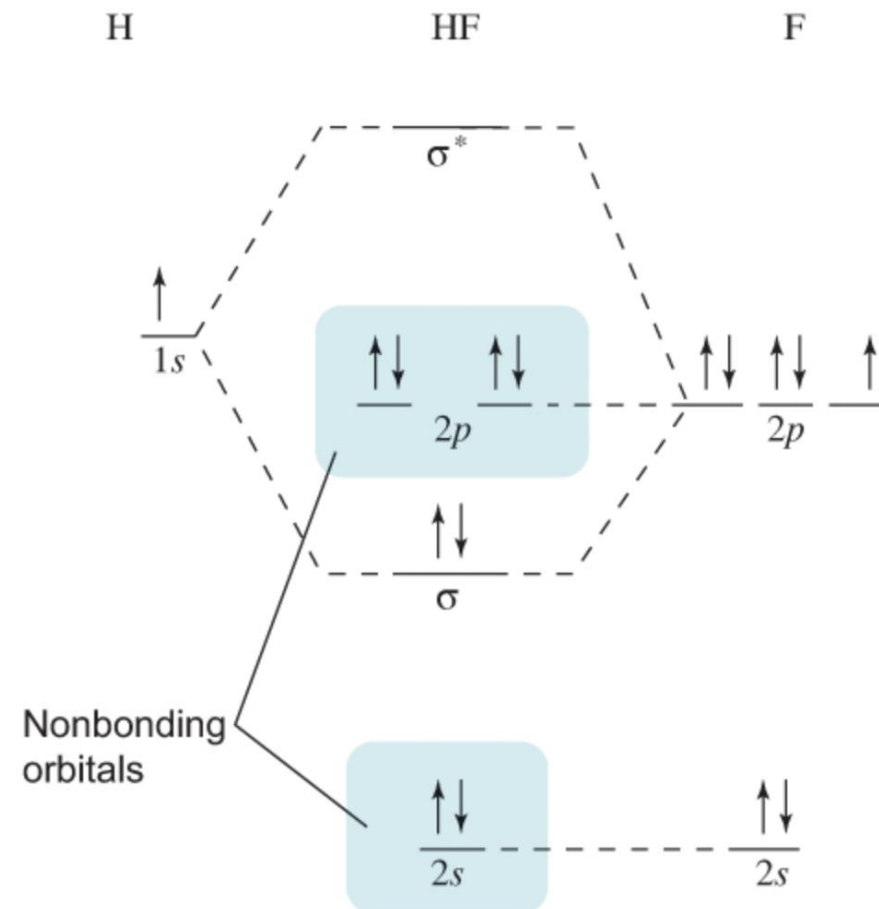


Figure 11.44.

11.5 Nonbonding Orbitals in H—X Molecules

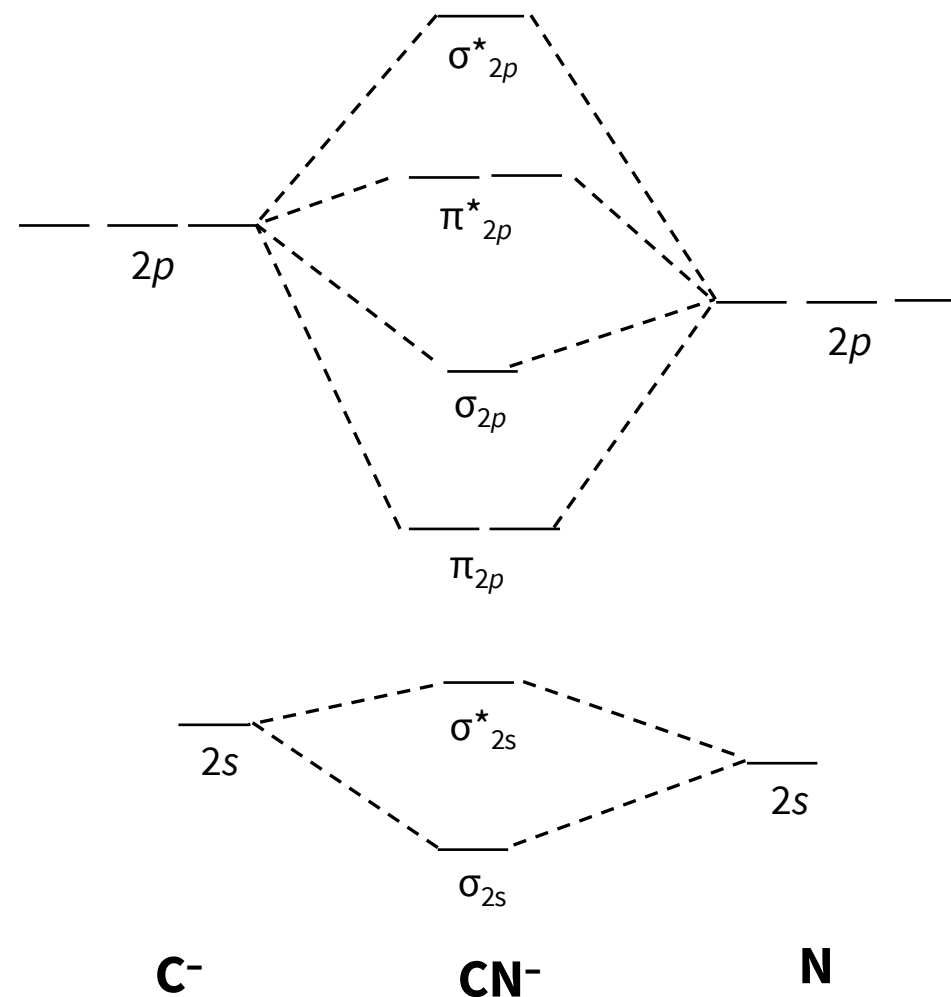
- Why are there three nonbonding orbitals in hydrogen fluoride?



11.5 Molecular Orbitals of Cyanide Anion

Example. Cyanide ion (CN^-)

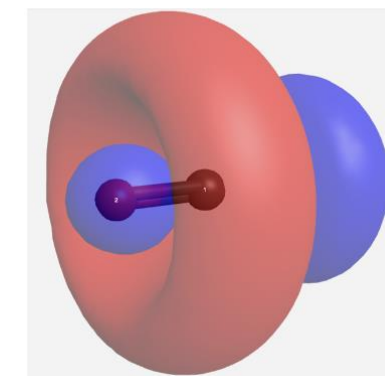
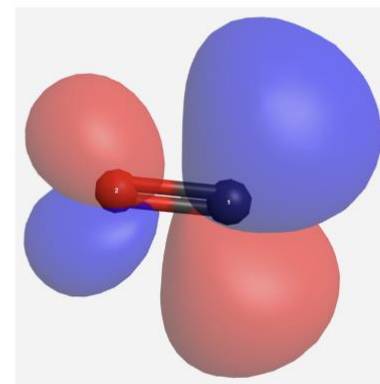
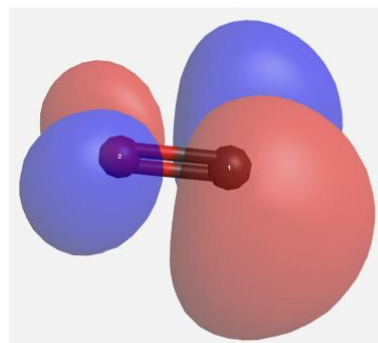
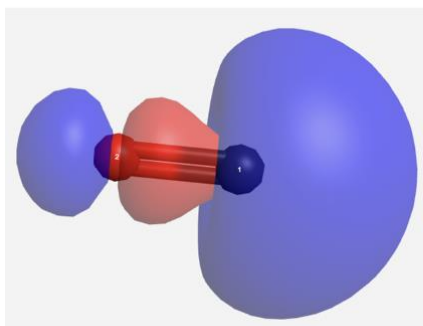
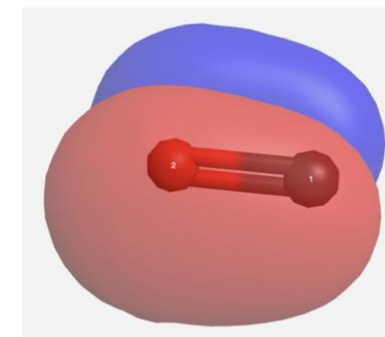
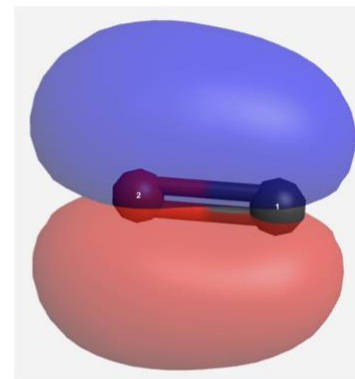
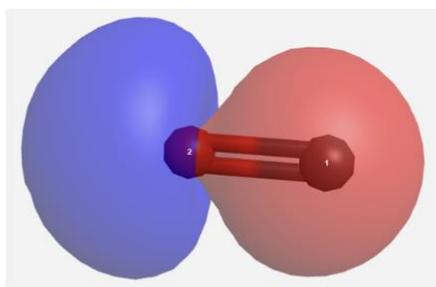
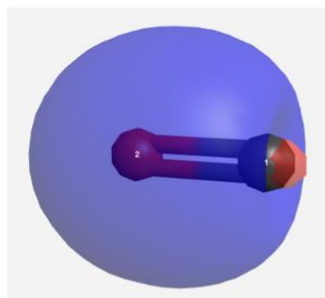
- Note the energies of the σ_{2p} and π_{2p} orbitals. s - p Mixing is occurring!
- Add electrons to the MO diagram and draw a Lewis structure for CN^- . Do the bond orders agree?



In the molecular orbital diagram of OH^- , the σ bonding orbital is formed from the $1s$ orbital of hydrogen with the $2p_z$ orbital of oxygen. This σ bonding orbital has a greater contribution from which of the following orbitals? For the purposes of this question, the internuclear axis is the z -axis.

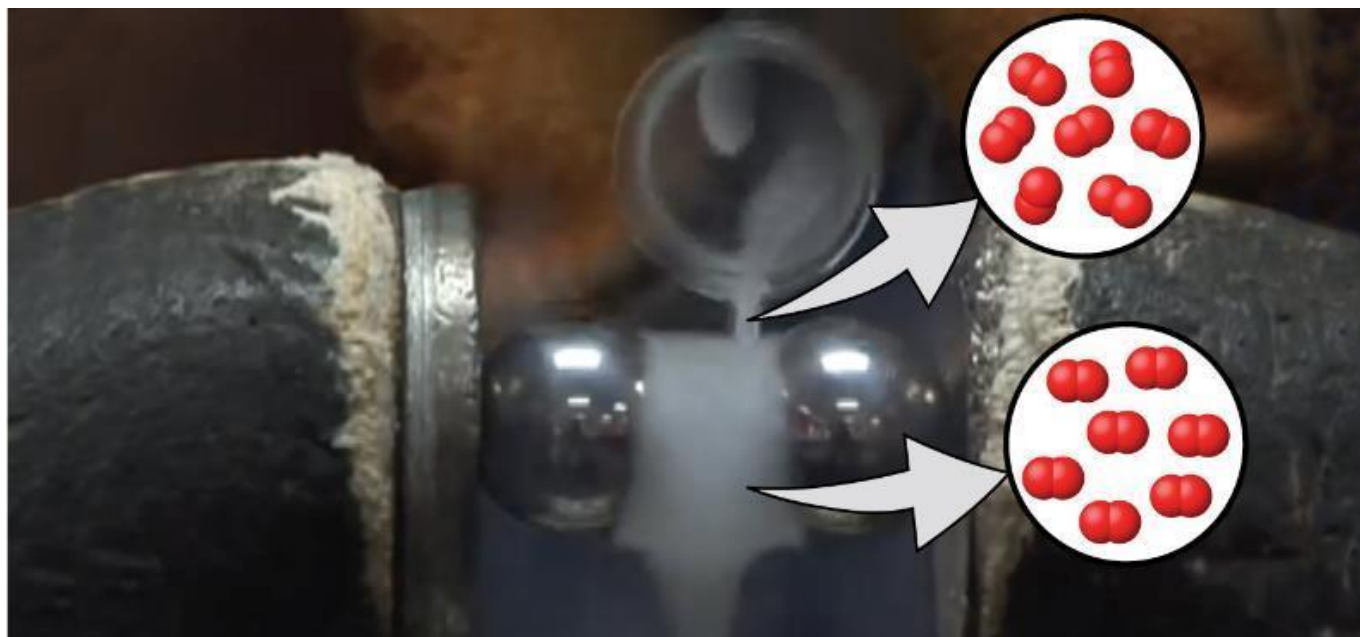
- A. $1s$ orbital of hydrogen
- B. $2p_z$ orbital of oxygen
- C. Both atomic orbitals contribute equally to this molecular orbital.

The molecular orbitals of CO are shown below from lowest to highest energy. Label each orbital as sigma or pi and bonding or antibonding. Atom 1 is carbon and atom 2 is oxygen. How many pi-type orbitals are shown?



11.5 Paramagnetism of Dioxygen

- Molecular orbital theory explains the **paramagnetism** of O_2 , while valence bond theory and Lewis structures do not.



OpenStax Chemistry 2e. Liquid O_2 is attracted to a magnetic field.

Determine whether each of the molecules below is paramagnetic or diamagnetic.
Consider the molecular electron configuration in each case to justify your conclusion.
Select the *paramagnetic* molecules.



11.5 Summary of Structural Theories of Molecules

Theory	Summary	Uses	Drawbacks
Lewis Theory	Atoms in molecules and polyatomic ions have completed valence shells following the octet/duet rule	Molecular diagrams show relative bond strengths as single, double, and triple bonds.	Two-dimensional model of three-dimensional molecules; no distinction between bonds in different compounds. Does not explain how/why bonds form.
VSEPR	The number of electron domains around the central atom in Lewis structure determines the three-dimensional shape of the molecule or polyatomic ion.	Predicts shapes and can apply polar bonds to determine the polarity of the molecule.	Does not explain how/why bonds form.
Valence Bond Theory	Covalent bonds form by overlap of atomic orbitals in sigma and pi bonds. The degree of overlap determines the strength of the bond.	Description of pi bonds helps to explain greater reactivity of compounds with double bonds and lack of rotation around double bonds. Hybrid atomic orbitals explain some molecular geometries, but not all.	Does not explain certain properties such as paramagnetism. Not easily used to predict properties of hypothetical molecules or ions.
Molecular Orbital Theory	Atomic orbitals combine to form bonding and antibonding molecular orbitals that extend over the entire molecule.	Explains the electron energy levels of molecules, explains paramagnetism/diamagnetism, and makes it possible to predict the properties of hypothetical molecules and ions.	Easily used for diatomic molecules of the first- and second-period atoms; becomes more complex as the size of the molecule increases.